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2024

The International Mathematical Modeling Challenge (IM^2C) Summary Sheet

Picking the Perfect Pet: A Household Readiness Model that Promotes Pet Ownership and Decreases Pet Abandonment

Would you like this one of a kind golden retriever pup? Or maybe this bobtail kitten! Maybe you're looking for something a little more... exotic. Hold on. Can you even take care of these pets? I'm going to need to see some credentials!

With the onset of the COVID-19 virus, the pet industry skyrocketed. People spent more time at home than ever, leaving them wanting the companionship pets can offer. However, how many people can actually support these pets?

That is our task today as mathematicians: to determine criteria for pet readiness and apply it to the real world, analyzing what factors are most important to consider to reduce abandonment and relocation of pets. Pets have psychological connections with their owners and relocation can be seen as betrayal and negatively impact their minds. Pets have feelings too, you know!

We approached the worthwhile questions posed in this challenge with a variety of mathematical techniques. First was the development of our Hyper-Rectangular Space Partitioning with Linear Inequalities Technique (HR-SPLIT), which was the framework for our readiness model based on user inputs. These inputs were strategically chosen to capture the widest possible picture of the household with the intention of comparing it to our standards for pet readiness. These included house space, money willing to spend on the pet, free time, knowledge on a pet species, and distance to a vet. HR-SPLIT allowed us to determine ranges of readiness, creating a spectrum rather than binary. This way, for example, people can see even if they are Somewhat Prepared to own a pet, in which they may want to improve their conditions before getting one!

We then used this model to determine readiness percentages for three countries: the U.S., U.K., and Japan. Using normal distributions, gamma distributions, and other probability functions, we were able to fit data we found to characterize these entire countries. As our final results, we found that 29.2% of households in the U.S., 41.2% of households in the U.K., and 36.2% of households in Japan are ready for pet cats.

Then, we extended our model to include four new species along with cat: dog, parakeet, goldfish, and red-tail boa. We also added on to which factors our model considers, adding on prior pet ownership as well as long term commitment factors to see if a household can support a pet for its whole lifetime. We discussed the impact of each of these factors through multiple household examples in New York, U.S.

Next, we were tasked with creating a model to predict pet household demographics and retention rates. For this, we used stochastic differential equations to implement a factor of probability, since the future and its events are truly unpredictable. We also used a lifetime density function to accurately determine the probability of death for pets. These two methods combined allowed us to determine household number projections for each of the countries previously considered as well as every species in 5, 10, and 15 years.

All of our final results are included at the end of this paper. For households with pet cats in the U.S., there are currently 46.5 million as of 2024. Our model projects that there will be 50.7 million households by 2029, 55.3 million households by 2034, and 59.64 million households by 2039 with pet cats. Of these, we determined a 95.3% retention rate, meaning that 4.7% of households that had pets the year before abandoned or surrendered their pet. For households with pet dogs in the U.S., there are currently 65.5 million, and our model projects that there will be 72.54 million by 2029, 80.01 million by 2034, and 87.92 million by 2039. Though these numbers are high, this comes with a 88.8% retention rate, which means that we are predicting 11.2% of households with pet dogs to surrender them! Our readiness model raises awareness about the plight of these pets and can hopefully spark meaningful discussions about their future.

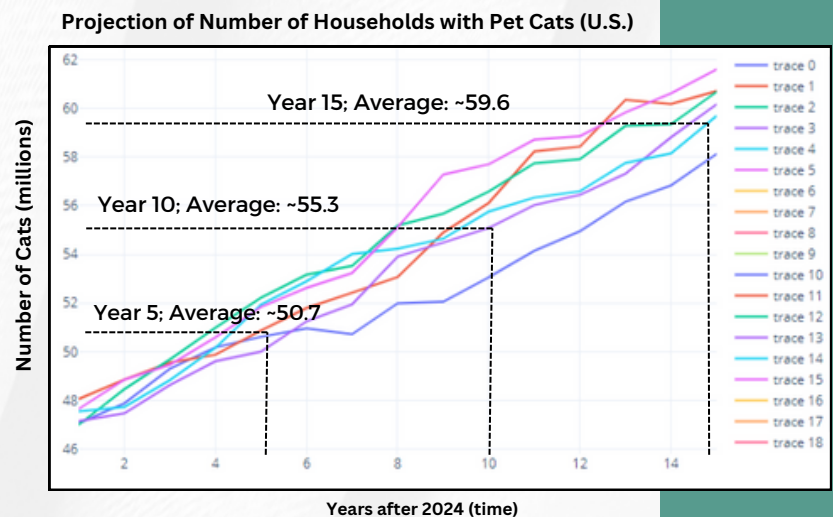
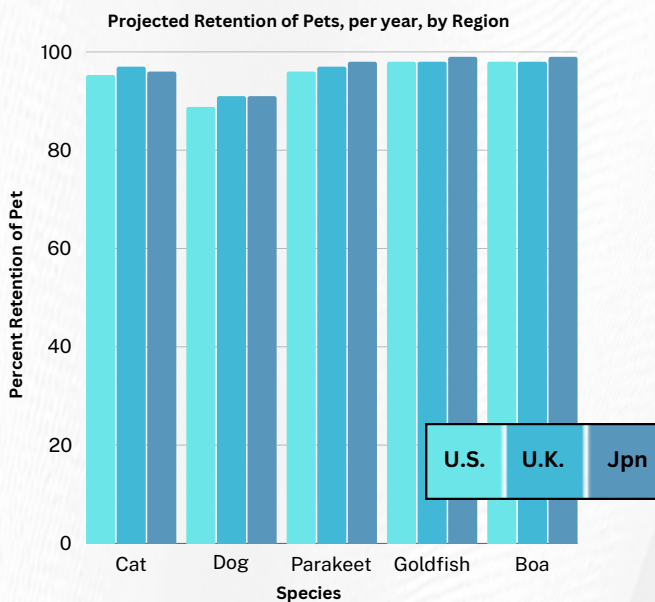
LETTER TO IMMC-A DIRECTORS

Dear Decision Makers,

Our team has conducted careful research on the pressing issue of pet abandonment, especially in the post-pandemic era, and we are excited to share that we have arrived at a final solution.

The model accounts for six unique factors that each have a significant impact on determining if a household is ready for a pet: physical restrictions, time constraints, potential financial stress, the health of the pet, and the knowledge that the potential owner has about the pet.

By using techniques such as HR-SPLIT (Hyper-Rectangular Space Partitioning with Linear Inequalities Technique), stochastic differential equations, probability density functions, and decision trees to model the flow of data, we have created what we believe to be accurate projections of pet ownership and retention in multiple countries for several pets.



This recommendation model balances a multitude of factors that give a holistic view of pet ownership that allows us to effectively measure pet retention and the projection of households with pets. These models allow us to see differences in retention rates between different types of pets. These inequalities give rise to more questions: Why do dogs have the lowest retention rates? What allows pet owners in the U.K. and Japan to have higher retention rates than the U.S., and how can Americans implement those measures? Such questions must be researched further, but our models provide a meaningful starting point.

It is our hope that our model may be used as a catalyst for more of these types of questions and research for the benefit of our beloved pets.

Thank you for your trust in us.

Best,
Team US-14839

For over 10 years, our team has been improving the lives and conditions of our pets and their owners. We are proud to be a part of this journey.

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1 Introduction

“The only, absolute and best friend a man has, in this selfish world, the only one that will not betray or deny him, is his dog.” —King Frederick II of Prussia, 1768

Dogs have been domesticated for at least 30,000 years, and cats for at least 12,000.^{[1][2]} Within the context of the long history of coexistence between humans and domesticated animals, it may come as a surprise that pets, in the modern sense, have only existed since the eighteenth century.^[1]

In the few centuries since then, pets have become major components of billions of people’s lives. In the U.S., there are more than twice as many dogs and cats as there are children.^[1] Internationally, the global majority of families own pets, and this majority is only growing.^[3] This growth has skyrocketed in the past several years, with almost one in five American households adopting a pet during the COVID-19 pandemic.^[4]

It is not difficult to understand the popularity of pets, especially during the pandemic. The sense of companionship they offer also provides a myriad of tangible health benefits.

Unfortunately, the sobering reality is that a sizable percentage of new pet owners learn too late that they are unprepared for the commitment of providing for a pet: in the U.S. alone, 6.3 million pets enter animal shelters annually.^[5] Thus, it is more important than ever to inform potential pet owners about the temporal, financial, and physical commitment that comes with living with a pet. The goal of our model is to help them navigate this decision.

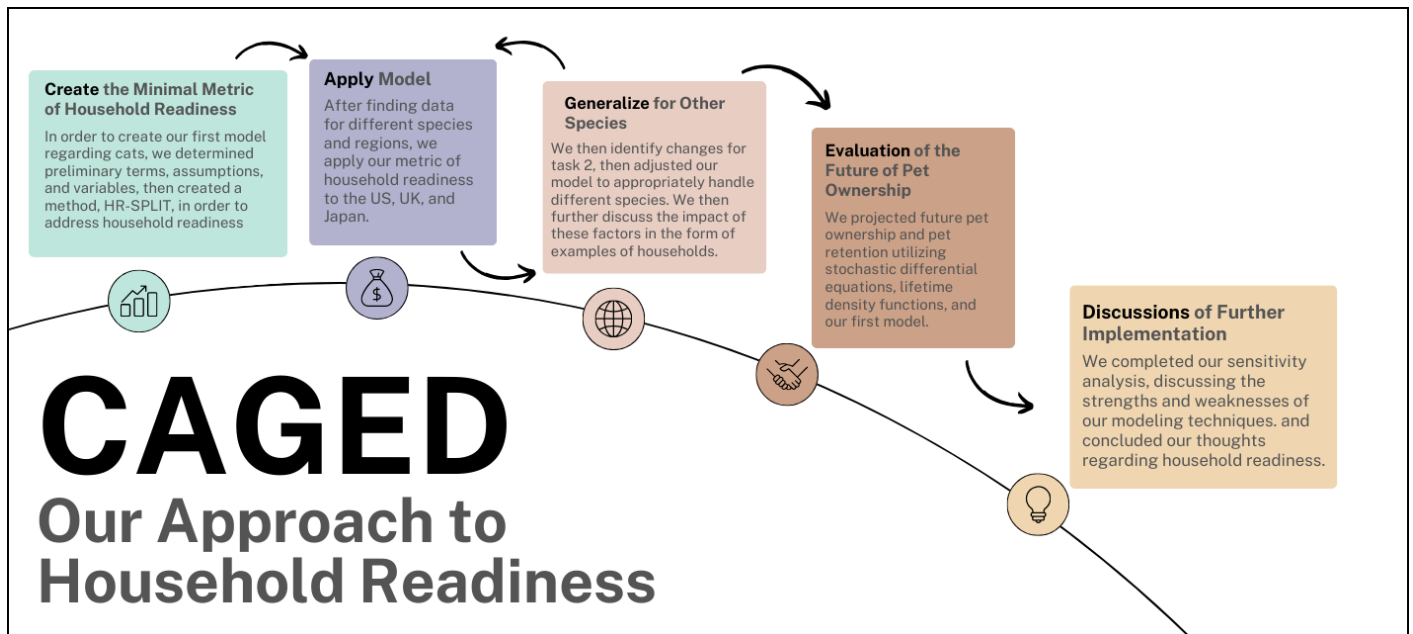


Figure 1: Flow Chart Process

1.1 Problem Restatement

1. (a) Create a mathematical model that can be used to determine whether a household is ready for a pet cat. This model takes in at most 10 pieces of user-inputted data. Justify these factors and what role each factor plays in determining the readiness of a household. Then, apply our model to three households that fit our “cat-ready” requirements in New York, U.S., and at least three households that do not fit our criteria.
- (b) Next, we use our model to fully describe the readiness of three countries of our choice: the U.S., the U.K., and Japan. This entails gathering sufficient data on these countries to get a distribution of readiness and find the number of households that are ready for a pet cat.
2. (a) Generalize our model from part 1 to four additional species of our choosing: dogs, parakeets, goldfish, and red-tailed boas. This includes adding and changing factors considered in the original model to better capture readiness for each specific species. Then, analyze the readiness of at least six illustrative examples of houses in New York, U.S.
- (b) Discuss how households with multiple pets impact the way our model computes readiness.
3. Create a model that predicts pet ownership and retention levels of the three countries considered in 1b for each of the five species used in 2a. Use the data captured in question 1b to determine the readiness of households and

represent potential successful pet ownership. Use our model to predict pet ownership and retention in 5, 10, and 15 years for each pet in each country.

4. Write a letter to the IMMC-A describing our models and how their results can be applied to the real world, allowing animal shelters or pet stores to reasonably decide if a household is ready for pet ownership or not. Provide insight on how using our model would decrease pet abandonment or mistreatment, contributing to the overall health of pets worldwide.

2 Preliminary Assumptions and Definitions

2.1 Term Definitions

Table 1. Term Definitions

Term	Definition
Pet	Merriam-Webster defines a pet as “a domesticated animal kept for pleasure rather than utility.” ^[6] Our definition slightly diverges from this; instead, we define a pet to be any domesticated animal not kept exclusively for utility. This eliminates potential confusion regarding pets that are kept for both pleasure and utility: for instance, therapy dogs are generally considered pets even though they also provide utility.
Utility	We say that an animal is kept for utility if they provide a service. Examples of domesticated animals kept exclusively for utility include cattle and police dogs.
Domesticated	We will use Merriam-Webster’s definition: “adapted over time [as by selective breeding] from a wild or natural state to life in close association with and to the benefit of humans.” ^[7]
Cat	We define a cat as a subspecies of the <i>Felis</i> genus known as <i>Felis catus</i> , which encompasses all breeds of domesticated cats.
Capable	We define a person living in a given household as capable if they are physically and mentally able. Generally, this excludes young children, elders, and disabled individuals.
Living Space (for pet)	The amount of space (m^2) and height (m) provided for the pet to live in.

2.2 Variable Definitions

Table 2. Variable Definitions

Variable	Definition
HRI	House Readiness Index
C_H	Annual amount of money the household is willing to dedicate to the pet
C_{pet}	Average annual cost of the pet
$C_{pet,Y/N}$	Average annual cost of the pet, potentially incorporating pet insurance
FS	Floor space (m^2)
FS_{pet}	Floor space required by the pet (m^2)
FS_H	Floor space in the household (m^2)
VS	Vertical space (m)
VS_{pet}	Vertical space required by the pet (m)
VS_H	Vertical space in the household (m)
FT	Free time

FT_{pet}	Average daily free time required by the pet (hr)
FT_H	Average daily free time of those in the household (hr) (among all capable members)
T	Type of location the household is in (urban, suburban, or rural)
VD	Distance between the household and the nearest veterinarian
MVD_T	Maximum allowed VD for the pet for a given T
IK	Inputted pet knowledge level

2.3 Assumptions

1. We assume that the people in the household want the pet.

Justification: If the people in the household do not want the pet in the first place, they would not get a pet even if their house were ready for one. So, we only consider households that want a pet.

2. We assume that allergies are of minimal concern due to the existence of hypoallergenic breeds, which are bred specifically to minimize allergic reactions for individuals typically sensitive to cats.

Justification: Advancements in breeding cats have led to the creation of hypoallergenic breeds, allowing households with allergy-prone members to still consider cat ownership. This expands the pool of potential pet owners and ensures that the presence of allergies does not automatically disqualify a household from cat ownership.

3. We assume that the households are aware of and ready to comply with pet policies imposed on their living conditions

Justification: Legal restrictions or apartment policy can affect a household's eligibility to own pets, and this assumption ensures that the household is able to get and host the pet in the first place.

4. We assume that the user input is truthful regarding the household and its members.

Justification: Our models cannot operate on false data, so we assume that the user input is truthful and accurately represents the household that the pet would live in.

3 Task 1: Everyone Wants a Cat

3.1 Factors Taken From the User

Table 3. Factors Considered

Factor	Description & Justification
Floor space in the household in m^2 (FS_H)	Cats are adaptable and can thrive in small spaces, but they still need room to move, play, eat, and sleep. A certain minimum amount of floor space is required for cats.
Vertical space in the household in m (VS_H)	Cats enjoy climbing and exploring vertical environments. A household that can provide such an environment may be more suited to cat ownership, as long as the horizontal square footage requirement is also met.
Annual amount of money the household is willing to dedicate to the pet in USD (C_H)	Financial readiness to own a cat encompasses the ability to afford ongoing expenses such as food, litter, toys, veterinary care, grooming, and pet insurance. Additionally, other initial costs such as spaying or neutering may be recommended.

Average daily free time of those in the household in hours (among all members) (FT_H)	Time availability is crucial for providing cats with the necessary care, attention, and social interaction they require. Dedicated time is needed for daily care tasks such as feeding, cleaning litter boxes, grooming, and veterinary visits. A household with more available time may be better positioned to meet a cat's needs.
Location of the household (T)	A choice of urban, suburban, or rural. It is important to know the environment the household is in.
Distance between the household and the nearest veterinarian (VD)	Ready access to quality medical care for the pet is important for regular checkups and especially for medical emergencies. So, a household will benefit from being closer to a Veterinarian.
Pet knowledge (IK)	The household must be knowledgeable on how to take care of pets and aware of specific dangers. The user inputs a value of 0-10 of their current knowledge or to what level they are willing to learn what is required to take care of their potential cat.

3.2 Development of Model

We proposed several unique methods to tackle this problem. Initially, we wanted to weigh each of the factors in proportion to their effect on household readiness. One suggestion was to use machine learning algorithms, but such a process would require a surplus of data that is impossible to gather. Another idea was to use expected values to characterize the readiness of all households in America and use them to estimate where the inputted statistics fell using a decision tree. This required data specific to each country and made it difficult to make conclusions about a given household's free time or the money they were willing to spend.

We arrived at a linear programming approach. Linear programming uses a system of linear inequalities to determine a feasible area that satisfies those constraints, then looks through that feasible region to optimize some objective function. In our case, instead of optimizing an objective function, we want to find feasible regions that indicate that a household is ready for a cat.

3.2.1 The Hyper-Rectangular Space Partitioning with Linear Inequalities Technique (HR-SPLIT)

Each of our factors acts as a constraint on the population of houses that are cat-ready. Given these constraints, we develop a system of linear inequalities to determine cat-ready households. However, instead of making each inequality binary, as they traditionally are, our model places value on inputs that are well above the minimum requirement, as this would represent a higher quality household for the pet. We implement a system of linear inequalities for each factor, where values that are higher than the requirement contribute more to the household's Readiness Index.

Each linear inequality partitions space and each region from this partitioning represents a different level of readiness for each factor. We partition a dimension of space for each factor considered, visually splitting space into hyperrectangles. We have assigned values to each of the individual partitions and take the geometric mean of these factors to obtain the HRI . The reason that the geometric mean is chosen is that it preserves normalization, so a HRI of 0.5 remains the expected center of distribution. HR-SPLIT is visually demonstrated in Figure 2 for one-dimensional, two-dimensional, and three-dimensional spaces. In the case of our model for cats, we have six constraints, so this model is extended into six dimensions. Each region has a value obtained by the geometric mean of six elements. It is impossible to visualize six-dimensional space, but Figure 2 allows for some conceptualization of the partitioning of space. We use this method for our model to determine household readiness for pet cats.

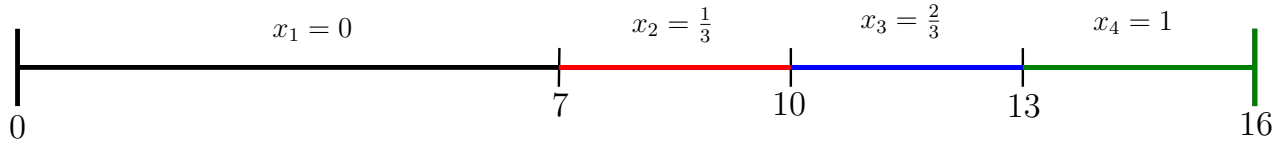
3.2.2 Implementation of Factors

Cat Cost

A general initial year cost for cats is \$1904, with annual costs after that averaging \$1149^[10]. A pet emergency will occur at least once in its lifetime, incurring a bill of \$2000 to \$4000^[11]. We take a conservative estimate of one \$3000 emergency in our cat's lifetime, which is about 15 years for an indoor pet cat.

Pet insurances cover major emergency costs, such as treatment for accidents and illnesses, but they pose complications for our model. Some jobs offer pet insurance as job benefits; otherwise, the household can choose to pay for insurance annually even if their pet is generally healthy. We will calculate a cost factor both with insurance and without insurance.

First, we will calculate the cost of providing for a cat without buying pet insurance. For an average year, it costs \$1149 to provide for a cat. In the first year, it costs an additional \$755. Using the assumption that the cat will suffer a



(a) 1D space partitioned with example values



(b) 2D space partitioned. The value assigned to each region is the GM of the individual partition values on each axis.

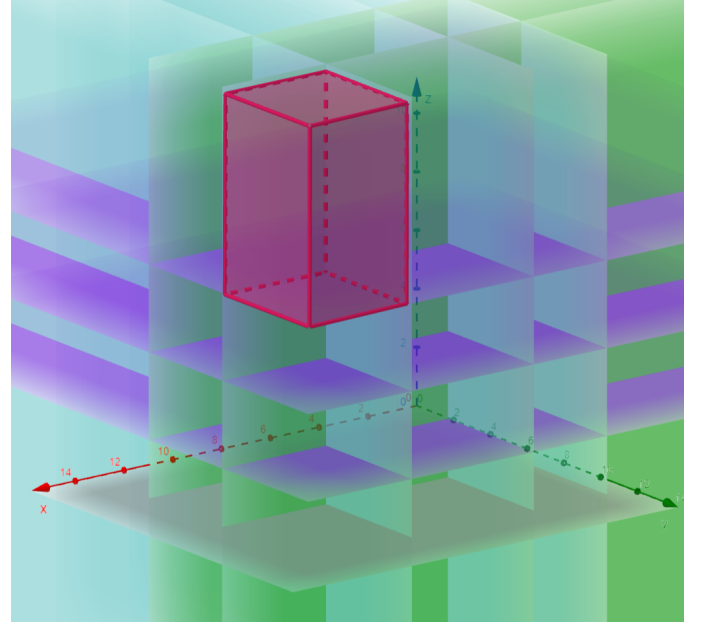
(c) 3D space partitioned. The red prism represents the region where $\sqrt[3]{x_i y_j z_k} = 1$, which is when $x_i = y_j = z_k = 1$.

Figure 2: HR-SPLIT visualized in 1, 2, and 3 dimensions

single emergency that costs \$3000 in one of its 15 years, we can calculate the expected no-insurance average to be

$$C_{cat,N} = \$1149 + \$755 \left(\frac{1}{15} \right) + \$3000 \left(\frac{1}{15} \right) = \$1399.$$

Assuming that the pet insurance covers all emergencies, we can expect an annual insurance cost (in the U.S.) of \$387^[13]. We find that

$$C_{cat,Y} = \$1149 + \$755 \left(\frac{1}{15} \right) + \$387 = \$1586.$$

Though it is more expensive to pay for pet insurance on average, we may recommend that households with high C_H buy pet insurance in the case of a medical emergency. In the general model, we incorporate the no-insurance cost as the minimum and check if the household can comfortably afford insurance if they are pet-ready.

In our system of linear inequalities for price, we assign values depending on the difference between C_H and C_{pet} . We have

$$f_C(C_H, C_{pet}) = \begin{cases} 1 & 2C_{pet} \leq C_H \\ \frac{i}{10} & (1 + \frac{i-1}{9})C_{pet} \leq C_H < (1 + \frac{i}{9})C_{pet} \\ 0 & C_H < C_{pet} \end{cases},$$

where i is an integer between 1 and 9, inclusive. In this function, we set the upper limit to be 2 times the minimum cost to provide for the cat, and we split the remaining favorable region into 10 subregions. If a household is willing to spend more than twice the minimum annually to take care of the cat, they will be able to comfortably provide quality food and toys and cover emergency medical expenses. We will develop a function like this for each factor and combine them to produce our final model.

Living Space, Floor Area, and Vertical

We then ask for the floor area and vertical space the household can provide for the cat. The absolute minimum space requirement for cats, as described by the Official Journal of the European Union^[8], is about $2 m^2$ of floor space (including shelf space) and $2 m$ of vertical height. However, it is known that the quality environment and the extent to which the cats are able to climb and explore should take precedent over the physical size of the space.^[12] We have already accounted for the quality of the space in the money factor (for instance, a household with a greater C_H will be more likely to buy a cat shelf), so the upper bound for these two factors will not be as high. Specifications for both floor space and vertical space functions are provided in Appendix B.

Free Time

In addition to at least 30 minutes of undivided attention to the cat every day, there must be enough time to take care of the cat. This includes time for feeding, watering, cleaning the litter box, and other general maintenance. Regular feeding, watering, and cleaning averages 30 minutes per day. To account for other maintenance activities, as well as extra quality time spent with the cat, we choose a minimum time investment of 1.25 hours. So, we have $FT_{cat} = 1.25$ hours. Then,

$$f_{FT}(FT_H, FT_{pet}) = \begin{cases} 1 & 5FT_{pet} \leq FT_H \\ \frac{i}{5} & iFT_{pet} \leq FT_H < (i+1)FT_{pet} \\ 0 & FT_H < FT_{pet} \end{cases},$$

where i is an integer between 1 and 4, inclusive.

Distance to Veterinarian

In the case of an emergency, the household must be in driving distance from a veterinarian, with closer distances being more favorable. However, any limits we impose must depend on the location of the household. In an urban setting, a veterinarian is likely to be located closer than in a rural area, so we set different distance limits for each. This is because distance doesn't necessarily correlate to time spent traveling, since traffic is higher in urban than in rural locations. For rural, suburban, and urban settings, respectively, we set upper limits of $MVD_T = 15, 10,$ and 5 miles. The function $f_{Dist}(VD, MVD_T)$ is defined similarly to the previous functions, with specifics provided in Appendix B.

Pet Knowledge

Many pet owners learn about how to take care of their pet after the pet is acquired, but our model requires some initial knowledge to ensure that the household does not immediately endanger their pet. Due to this, we will only consider households with at least a 4 on the knowledge scale. If this section is the only one holding someone back, that user can be directed to resources where they can learn more about cat safety. We have

$$f_{know}(IK) = \begin{cases} 1 & 8 \leq IK \\ \frac{i}{5} & i+3 \leq IK < i+4 \\ 0 & IK < 4 \end{cases},$$

where i is an integer ranging from 1 to 4, inclusive.

3.2.3 Model 1

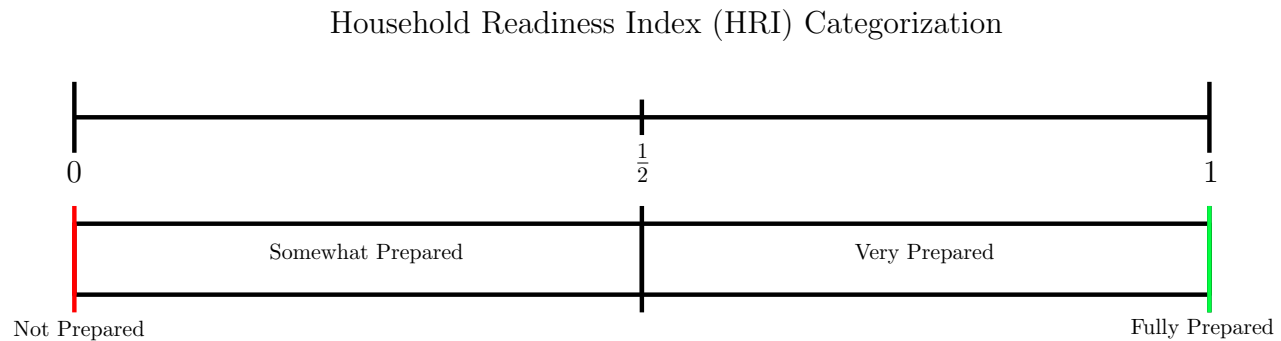
To combine each of these factors, we take the geometric mean of the outputs of f . Let $K = \{C, FS, VS, FT\}$ be the set of variables dependent on the type of species. Then, let $S = \{Dist, know\}$ be the set of variables independent of the type of species. The HR-SPLIT model to determine a house readiness for a pet cat is described by

$$HRI = \left(f_{know}(IK) \cdot f_{Dist}(VD, MVD_T) \cdot \prod_{k \in K} f_k(k_H, k_{cat}) \right)^{\frac{1}{|K \cup S|}}.$$

By taking the geometric mean, we ensure that if a household fails any of the criteria by achieving a score of 0, it will be deemed unsuitable to buy a pet cat. For each pet, our user-friendly model will output one of four categories that the user falls into based on their HRI: Not Prepared, Somewhat Prepared, Very Prepared, or Fully Prepared.

By using the geometric mean, we are able to preserve a normal distribution of scale in the final HRI. This means that we can give users a percent value of how close they are to being Fully Prepared for a pet cat. If we combined the

factors in another way, perhaps by multiplying them, the scale shown in Figure 3 would not be preserved. That is, if some household scored $\frac{1}{2}$ in every category, their HRI would be $\frac{1}{2}$, the theoretical median of scores.



3.3 Application of Model

3.3.1 Application of Model for (a)

In order to capture the diversity of households that could have a cat as a pet in New York, the United States, we apply our model to three households of varying locations which include urban, suburban, and rural. We base each example on data from the surrounding areas of the region.

Table 4. Data & Results of Very Qualified Households

	House 1	House 2	House 3
Location Type	Manhattan, NY (Urban)	Queens, NY (Suburban)	Suffolk County, NY (Rural)
C_H	\$2500	\$1900	\$4000
FS_H	20 square meters	75 square meters	140 square meters
VS_H	2.4 meters	2.5 meters	2.7 meters
FT_H	3 hours	3 hours	8 hours
VD	3 miles	7.8 miles	11 miles
IK	8	7	6
Result	0.6887	0.5267	0.7647

Each of these houses surpasses the minimum requirements for a pet cat, so they are all classified as Very Prepared ($HRI > 0.5$).

House 1 is a well-off apartment in Manhattan, with ample free time and money to look over a cat. They have also done research on what is expected of cat owners to ensure a pet cat remains healthy throughout its lifetime, so they have a high knowledge stat. House 2 is a spacious household in Queens, and the family has enough money and free time to take good care of a pet cat. House 3 is a wealthy rural household. The members of this household have a lot of free time and space for a cat to thrive in. In addition, our model recommends that Houses 1 and 3 get pet insurance due to their high C_H stat.

Table 5. Data & Results of Somewhat Qualified Households

	House 4	House 5	House 6
Location Type	Manhattan, NY (Urban)	Nassau County, NY (Suburban)	Amenia, NY (Rural)
C_H	\$2300	\$1500	\$1600
FS_H	18 square meters	70 square meters	110 square meters
VS_H	2 meters	2.6 meters	3 meters
FT_H	2 hours	5 hours	3 hours
VD	4 miles	8 miles	15 miles
IK	6	4	5
Result	0.4785	0.3724	0.4692

Each of these houses are classified as Somewhat prepared, meaning they met every minimum requirement for owning a pet cat but may need to improve in some categories. House 4 is a city apartment with ample funds and floor space, but could improve on vertical space, pet knowledge, and free time if they wanted to be Very Prepared. House 5 is a suburban house with ample space and free time for a pet cat, but fall short in distance to a veterinarian, pet knowledge, and money they are willing to spend. House 6 is a rural household with excess floor and vertical space to house a cat, but it falls short in their C_H and distance from a veterinarian. These households are all ready to house a cat if they really wanted to, but some might want to improve their stats before getting a pet cat based on their HRI score.

Table 6. Data & Results of Unqualified Households

	House 7	House 8	House 9
Location Type	Manhattan, NY (Urban)	Brooklyn, NY (Suburban)	Hamilton County, NY (Rural)
C_H	\$2000	\$1200	\$2000
FS_H	15 square meters	60 square meters	150 square meters
VS_H	2.3 meters	2.4 meters	2.8 meters
FT_H	1 hours	4 hours	3 hours
VD	2 miles	8 miles	23 miles
IK	7	6	8
Result	0	0	0

These examples emphasize the fact that every requirement must be met for the household to be considered ready to any extent to get a pet. House 7's members do not have enough free time to adequately take care of a pet cat. House 8's members do not have enough annual expendable income to provide for a cat. Finally, though House 9 is in a rural environment, it is still located too far away from a veterinarian to receive immediate medical care in the case of an emergency.

3.3.2 Application of Model for (b)

Table 7. Additional Definitions for Task 1b

Variable/Function	Definition
$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$	Standard normal probability density function
$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$	Standard normal cumulative distribution function
$f(x; \xi, \omega, \alpha) = \frac{2}{\omega} \phi\left(\frac{x-\xi}{\omega}\right) \cdot \Phi\left(\alpha\left(\frac{x-\xi}{\omega}\right)\right)$	Skewed Normal Probability Density Function
$\Gamma(n) = \int_0^\infty t^{n-1} \cdot e^{-t} dt$	The Gamma Function
$f(x; \mu; \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	Normal Distribution Function
$f(x; \mu; \alpha) = \frac{1}{x\alpha\sqrt{2\pi}} e^{-\frac{(\ln(x)-\mu)^2}{2\alpha^2}}$	Log-Norm Distribution Function
$f(x; \alpha, \xi, \omega) = \frac{1}{\Gamma(\alpha) \cdot \omega^\alpha} \cdot (x-\xi)^{\alpha-1} \cdot e^{-\frac{x-\xi}{\omega}}$	Gamma Distribution
ξ	Location parameter of distribution, indicated central tendency
ω	Scale parameter of distribution, affects spread of distribution
α	Shape parameter of distribution, affecting skew
C_{living}	Cost of Living
I	Income of a household
$L_{total}, L_{total,weekday}, L_{total,weekend}$	Average total leisure time per day, weekday, weekend
$L_{other}, L_{other,weekday}, L_{other,weekend}$	Average leisure time for activities excluding pet per day, weekday, weekend
L_{pet}	Average leisure time for pets per day

One large acknowledgment that must be made is that the countries we choose for this model are all first-world countries. The reasons for this decision are twofold.

First, there are simply more pet-related data in wealthier countries.^[14] This allows the model to come to more accurate conclusions.

Second, our model is more applicable to potential pet owners in wealthier countries. Buying and providing for pets is an expensive endeavor that requires a certain level of disposable income, and for this reason, pet ownership is less common in developing countries. However, we also acknowledge that as pet ownership becomes more common in developing countries, this model would be more accessible if it were expanded to include them.^[15]

We first applied our model to the United States. According to data from the United States American Housing Survey, more than 97.6% of American homes are larger than 500 ft², or about 46.4515 m².^[16] This means that even among American homes whose floor spaces are on the 3rd percentile, the minimum 2 m² required for a cat is less than 4.3% of the total floor space. It is reasonable to assume that, strictly considering spacial requirements and not if they would want the cat in the first place, nearly all American residents would be ready to dedicate this space for a cat. Also, the International Code Council requires all habitable spaces to have a ceiling height of at least 2.286 m, which means that all legal living spaces are able to accommodate the minimum vertical space of 2 m required for cats.^[17] So, for the rest of this section, we do not have to consider the physical requirements for owning a cat.

We then considered the cost factor, which considers the number of people who are able to spend the necessary amount of money for the cat. (US Census income and Forbes cost of living) We graphed cost of living data of each state from Forbes on a histogram and found that the data was right-skewed. After applying a normal fit, a skewed normal fit, and a log-normal fit to match the data as shown in Figure 4a, we chose the fit with the highest r^2 value, which was the skewed normal fit that accounted for 80.1% of variation. We used the skewed normal probability density function with $\xi = 32374.44$, $\omega = 9728.09$, and $\alpha = 7.63$.

After obtaining the probability density function for the cost of living, we calculated the percentage of each income group that is financially ready for a cat. If $I - C_{cat} < C_{living}$ for a group, they are not ready. So, for each group, we computed the skewed cumulative distribution above $I - C_{cat}$ and multiplied by 100%. The end result of this initial

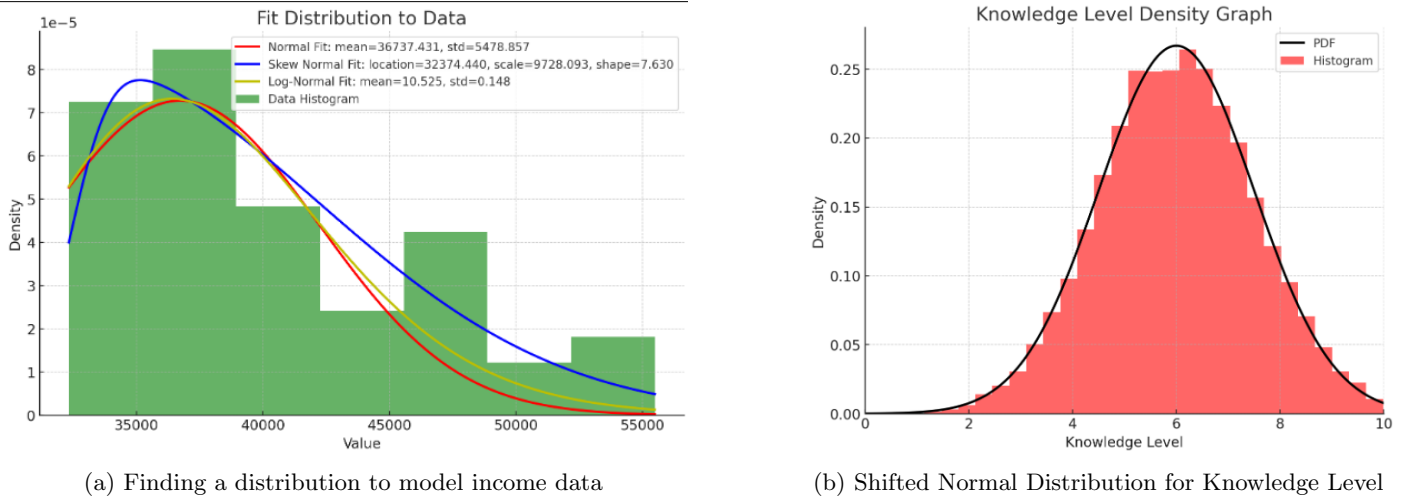


Figure 4: Distributions used to fit data

calculation for all income groups is provided in Appendix C. Generally, all groups under \$30,000 per year are not qualified, with qualification percentage increasing up to 100% of households at the \$60,000 range.

Next, we found data on the relationship between weekly salary (divided into ranges of \$0 to \$760, \$761 to \$1160, \$1161 to \$1810, and \$1811 and higher) and total leisure time.^[18] This leisure time was separated into weekends and weekdays and included activities like participating in sports, recreational activities, exercising, socializing, communicating, and watching TV. We assumed that the amount of time the household was willing to dedicate to the pet was

$$L_{pet} = L_{total} - L_{other},$$

where L_{other} is equal to the sum of the time spent on the leisure activities listed above. For each salary range, we computed L_{total} as follows:

$$L_{total} = \frac{5(L_{total,weekday}) + 2(L_{total,weekend})}{7}.$$

We computed L_{other} as follows:

$$L_{other} = \frac{5(L_{other,weekday}) + 2(L_{other,weekend})}{7}.$$

Now, we have L_{pet} for each salary range. Because this value is a mean, we assumed that the L_{pet} followed a normal distribution with a standard deviation of 1. With the distribution curves, we then calculated the percentage of each salary range that is ready for a cat by computing $1 - \text{cdf}(0, 1.25)$ of the distribution. All intermediate results are provided in Appendix C. We are left with roughly 43,732,817 households after considering the temporal and financial factors.

We then assumed that the distribution of knowledge levels of all the households followed a normal distribution. Because cats are so commonplace that even those who do not have cats are likely to have friends or family that do, it is reasonable to assume that the general public has an above-average knowledge of how to take care of a cat compared to other animals. So, we used a truncated normal curve centered at a knowledge level of 6 with a standard deviation of 1.5. We eliminated all households with $IK < 4$, as we deemed them unready in our model. We computed the percentage of households that are financially ready as approximately 90.85%.

So, from the total number of households we had remaining after eliminating households that were temporally and financially unready, we multiplied by 90.85%. $43,732,817 \cdot 90.85\% \approx 39,731,264$.

Then, we moved on to the distance from veterinarians.^[19] We first plotted the urban distance data on a histogram and found that it was approximately normal (a normal distribution fit gave an r^2 value of 0.961). The probability density function can be defined by the normal distribution function $f(x; \mu; \sigma)$ where $\mu = 4.22$ and $\omega = 0.06$. Using this distribution, we then found that all households located in an urban area were within 5 miles of a veterinarian and were thus suitable from our model.

We then analyzed the suburban data. We plotted the histogram and used graphing technology to find that the distribution for this data was best modeled by the gamma distribution (in Table 7) with $\alpha = 1.42$, $\xi = 7.68$, and $\omega = 0.71$. This distribution gave a r^2 value of 0.9638 so it is a good fit for our data. Now using this distribution, we compute a

$$1 - \text{cdf}(0, 10) \cdot 100\% \approx 7.88\%$$

Lastly, for rural data, we came across a problem: there was not enough data compared to the urban and suburban environments, and when we tried to find a distribution that fit, none fit. Thus, we assumed that the distribution was normal like the others. The distribution's probability density function can be defined with the normal distribution function $f(x; \mu; \sigma)$ where $\mu = 10.9569$ and $\omega = 0.386$. Using this distribution, we calculated

$$1 - \text{cdf}(0, 15) \cdot 100\% \approx 0\%$$

So, according to this model, all households located in a rural area are also within 15 miles of a veterinarian.

Because all households in urban and rural areas met the veterinarian distance requirement, we only needed to subtract the households that do not satisfy the requirement in the suburban area. Sixty-nine percent of the U.S. is suburban,^[20] so the final number of households that are ready for a cat is

$$39,731,264 - 39,731,264(69\% \cdot 7.88\%) \approx 37,570,995.$$

Thus, around 3.7571×10^7 households, or around 29.237% of U.S. households, is ready to own a cat.

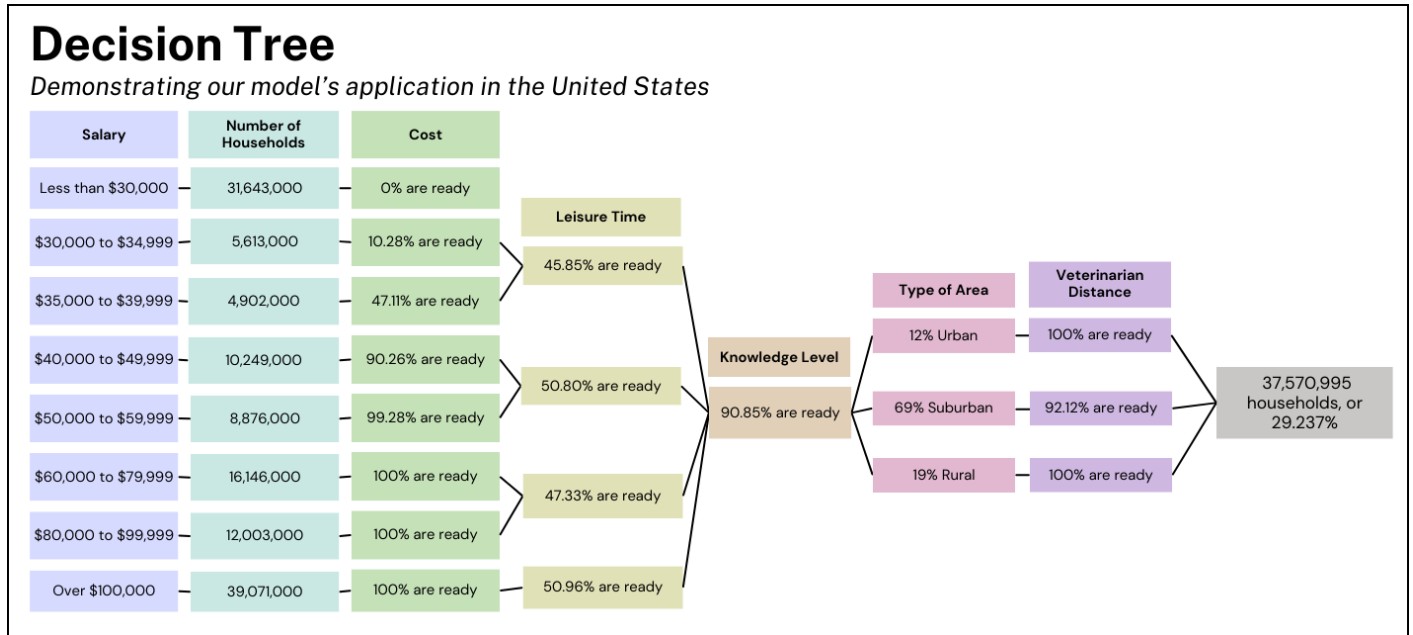


Figure 5: Decision Tree applied to the United States

Steps taken to determine the readiness of the U.S. were taken identically for the U.K and Japan, just fitting different data. All tables, diagrams, and decision trees for U.K. and Japan readiness can be found in Appendix C. The final readiness percentages are as follows: 36.16% of households in the U.K. are ready for a pet cat and 41.18% of the households in Japan are ready for a pet cat.

4 Task 2: Generalizing Our Model

When extending Model 1 to other species, it's important to recognize that while we have not changed our preliminary factors when applied to cats, many of the pet-dependent constants will change relative to the pet which we are extending our model to: dogs, birds, fish, and snakes. The region will also inevitably have an impact because of the economic factors and resources available in different locations.

Table 8. New Factors Considered

Factor	Description & Justification
Type of Pet (P)	The type of pet we are determining readiness for will allow us to extend our model to different species. Albeit this is still limited as there are hundreds of types of pet and our model will only be able to handle 5, but this will allow us to adjust our pet-dependent constants so that we can provide accurate household readiness data.
Prior Pet Ownership	In order to create a more versatile model and address the concerns of Task 2b, we are considering whether the person utilizing our model already owns a pet. We will prompt the user: If you have any other pets, how well do they get along with P ? This will allow us to give a more dynamic view of household readiness based on pet synergies and how they interact.
Long Term Commitment	This factor is considered in order to address social concerns of pet abandonment and the likeliness that a pet will be abandoned if the owner isn't aware or isn't willing to commit their time and resources to their pet. This gives us a more holistic view of who the prospective pet owners are and their willingness to support their pet.

Table 9. New Variable Definitions

Variable	Definition
P	Type of Pet
PC	Pet Cooperation statistic (default of 1 if no other pets)
mA_{pet}	Max Age of a pet species when in captivity
TE_H (years)	Commitment Time Expected of the household
TE_{pet} (years)	Pet species Age Expectancy

4.1 Impact of Additional Factors to Model

4.1.1 Prior Pet Ownership

The model prompts the user: If they have other pets, how well their pets get along with pet type P . Their answer will be on a range from 0-10, 10 being very well and 0 being horrible. Users also have the option to select that they do not have other pets, and in that case $f_{Opet}(PC) = 1$. Otherwise,

$$f_{Opet}(PC) = \begin{cases} 1 & PC \geq 8 \\ \frac{i}{5} & i + 3 \leq PC < i + 4 \\ 0 & PC < 4 \end{cases}$$

Where i ranges from 1 to 4, inclusive. This allows the model to incorporate the possibility that existing pets do not get along well with the desired type of pet, something that all households should make sure to do before getting a new pet.

4.1.2 Long Term Commitment

Different species of pets have different life expectancies, so some households may be better suited to different species based on the time commitment they are willing to give towards caring for one. The user will be able to input some number of years they would want to have the pet. The closer TE_H is to TE_{pet} , the higher $f_{TE}(TE_H, TE_{pet})$ will be. If $TE_H = TE_{pet}$ or exceeds it, $f_{TE}(TE_H, TE_{pet})$ returns 1. This is because it is possible to get a pet at virtually any stage of its life, and the closer a household's time commitment is to the full age expectancy of the pet, a larger age range for their

pet is possible. At the same time, committing to the entire pet’s lifetime isn’t required since pets can be adopted a few years into its life. We have

$$f_{TE}(TE_H, TE_{pet}) = \begin{cases} 1 & TE_H \geq TE_{pet} \\ \frac{i}{4} & \frac{i}{4}TE_{pet} \leq TE_H < \frac{i+1}{4}TE_{pet} \\ 0 & TE_H < 0.25TE_{pet} \end{cases}$$

Where i ranges from 1 to 3, inclusive.

4.1.3 Complete Readiness Model

With new factors, we now have $K = \{C, FS, VS, FT, TE\}$ and $S = \{Dist, know, Opet\}$. Then, our final HR-SPLIT model computes the Household Readiness Index as

$$HRI = \left(f_{know}(IK) \cdot f_{Dist}(VD, MVD_T) \cdot f_{Opet}(PC) \cdot \prod_{k \in K} f_k(k_H, k_{cat}) \right)^{\frac{1}{|K \cup S|}}.$$

4.2 Application to New Species (Task 2a)

Table 10. Species Data

	Cat	Dog	Parakeet	Goldfish	Red-Tail Boa
C_{pet}	\$1399	\$1762	\$264	\$374	\$520
FS_{pet}	$2m^2$	$6m^2$ [1]	$0.7m^2$ [11]	$0.25m^2$	$1.3m^2$
VS_{pet}	$2m$	$2m$ [1]	$0.5m$ [11]	$0.5m$	$0.7m$
FT_{pet}	1.25 hours	2 hours	1 hour	0.5 hours	0.5 hours
TE_{pet}	15 years	13 years [9]	15 years [10]	20 years	25 years

*Specific calculations for each factor are similar to that for the cat and are included in Appendix D. Otherwise, stats are taken directly from sources.

We decided to apply our updated model to ten houses in New York, U.S., to highlight the differences in qualification for each species individually as shown in Table 11 and Table 12. The category each score falls into is also abbreviated next to the HRI . Specifically, SP = Somewhat Prepared and VP = Very Prepared.

Table 11. Qualified Houses

	House 1	House 2	House 3	House 4	House 5
P	Cat	Dog	Parakeet	Goldfish	Red-Tail Boa
T	Harlem (Urban)	Nassau (Suburban)	Albany (Urban)	Bayside (Suburb)	Canandaigua (Rural)
C_H	\$3925	\$2100	\$800	\$1600	\$1900
FS_H	$10 m^2$	$15 m^2$	$0.95 m^2$	$1 m^2$	$5 m^2$
VS_H	$2.5 m$	$2.8 m$	$1 m$	$0.75 m$	$1 m$

FT_H	3 hours	3.5 hours	1 hour	1.5 hours	6 hours
VD	1.5 miles	9 miles	4 miles	5 miles	14 miles
IK	5	9	4	10	6
TE_H	15 years	20 years	12 years	15 years	20 years
PC	N/A	7	N/A	9	6
Result (HRI)	0.6932, VP	0.4942, SP	0.4902, SP	0.8603, VP	0.6787, VP

House 1 is a wealthy urban apartment with no other pets. Though it falls slightly short in pet knowledge and free time, it exceeds the minimum for a cat and the excess funds and space make this household very prepared for a cat. Further, our model also recommends this household gets pet insurance due to the amount they are willing to pay.

House 2 is a suburban family house with ample space and a cat that is generally friendly towards dogs. They have been wanting a pet dog for a while, so the family has studied on how to take care of it and is committed to have the dog for a full lifespan. They were rated as Somewhat Prepared by our model, but due to their C_H stat our model does not recommend this family gets dog insurance. Based on the results of the model, this house could improve to Very Prepared if they were able to dedicate more money or time to take care of a dog.

House 3 is an apartment in Albany. Their apartment isn't the largest and they don't have the most free time or funds for a pet, but they wanted to see if they were prepared to get a pet Parakeet anyways. The space they would designate for a cage is $1m$ tall with a floor space of $1m^2$. Despite their lower stats, they are still somewhat prepared to house a parakeet due to its lower requirements to care for.

House 4 is a suburban household, looking to add a new goldfish to its existing collection of fish. It already has a fish tank with $1m^2$ floor space and $0.75m$ height with a few freshwater fish, and want to see if they are able to handle a new goldfish. Our model ranks this household as Very Prepared.

House 5 is a rural, agricultural household. The owners have an interest in exotic pets and want to know if they are ready for a snake. Their other animals wouldn't interact much with the snake, but they know the introduction of a snake wouldn't have a negative impact on their other animals. Our model rates this house Very Prepared due to the excess space, funds, and free time this household has.

Table 12. Unqualified Houses

	House 6	House 7	House 8	House 9	House 10
P	Cat	Dog	Parakeet	Goldfish	Red-Tail Boa
T	Manhattan (Urban)	Elmhurst (Suburban)	Schenectady County (Rural)	Jericho (Suburban)	Albany County (Rural)
C_H	\$1600	\$700	\$800	\$700	\$1320
FS_H	$7m^2$	$10 m^2$	$0.5 m^2$	$0.9 m^2$	$3 m^2$
VS_H	$2.2m$	$2 m$	$1.2 m$	$0.6 m$	$0.8 m$
FT_H	1.5 hours	5 hours	3 hours	1 hour	2 hours
VD	10 miles	7 miles	12 miles	4 miles	5 miles
IK	6	5	10	2	7
TE_H	12 years	10 years	20 years	4 years	20 years
PC	N/A	N/A	9	N/A	1
Result (HRI)	0	0	0	0	0

Each of these houses is an instructive example of what traits our model values in households to ensure they are pet-ready.

House 6 is an urban apartment that meets the minimum requirement for most factors, except they are located too far from a veterinarian to reasonably travel there in the event of an emergency. Our model would recommend that, if they are able to get closer to a vet, that this household also improves their funds and free time to be better prepared for a pet cat.

House 7 is a suburban household where one large family lives. Among all of the capable family members, they have a sufficient amount of free time for a dog, but this family falls short in the amount of money they can put towards taking care of a dog. Our model advises this household to hold off on getting a pet dog until they have the sufficient funds.

House 8 is a rural household of bird enthusiasts with an existing small bird sanctuary in the home. They know everything there is to know about birdkeeping and their pet birds are very friendly. However, this household has too many birds, such that they don't have enough space left in the sanctuary to to healthily fit a new parakeet to their collection. As such, our model deems this household as Not Prepared for a pet parakeet, and they should consider investing in a larger sanctuary if they want more birds.

House 9 is a suburban household wanting to see if their house is ready for a goldfish. Though it has sufficient space and funds able to allocate to taking care of the fish, the members of the household is relocating in 4 years, so they would not be able to keep the fish for a sufficient fraction of its lifetime. In addition, the inputted fish knowledge is below 4, so this household would need to do some research before considering getting a goldfish after they relocate.

House 10 is a rural household interested in an exotic snake pet. They have sufficient money, space, and free time for the snake, however, they also have a pet dog. On a visit to the zoo a few years ago, their dog started growling at the snake sanctuary and had to be pulled away. Because of this, the owners reasoned that their dog definitely wouldn't get along with a pet snake, and inputted a 1 for *PC*, deeming this household as Not Prepared for this pet.

4.3 How does our model address multiple pets? (Task 2b)

In our model, we introduce a new factor: Prior Pet Ownership (PPO)– added when we considered the restraints of Task 2. Defined in section 4.1.1, PPO is an index, ranging from 0 to 10, designed to measure the compatibility or synergy between pets. This approach allows pet owners to predict how well a new pet will blend with existing ones, eliminating the need for extensive analyses of each pets' interactions, behaviors, and the age they are introduced to one another. By condensing these dynamics into a straightforward scale, PPO offers an easy solution to understanding the nuances of pet ownership and interaction, minimizing the need for exhaustive data collection and an incredibly detailed assessment.

5 Task 3: The Future of Pet Ownership

5.1 Pet Ownership and Retention Projection

5.1.1 Pet Retention

First, we updated our definition of readiness to include an aspect of time. A household's readiness status is volatile, so we decided that readiness for a household is determined annually. A household can be ready one year and not ready the next, so determining readiness annually makes sense. In a time lens, we also say that if our model determines a household as not ready that year, then they will be forced to surrender or abandon that pet purely by not having sufficient resources to take care of it.

To determine pet retention over time, we make an additional assumption about pet ownership:

If a household wants a pet and is ready for a pet, then they get that pet.

The reasoning for this assumption is that it allows us to determine what percent of households who have pets are ready for that pet or not, allowing us to figure out yearly retention. We make the inherent assumption that everyone who has a pet wants a pet, which is reasonable since having a pet is a voluntary decision.

In 1b, we found the percentage of households that are ready for a cat given they want the pet for each country. From this, we found a 29.2% : 70.8% split in ready and not ready households who want cats in the U.S. However, among the set of households that have pets, it makes sense that a higher percentage of them are ready for a pet. Since we say that households that are not ready will surrender/abandon their pet that year, we want to find the percentage breakup of Ready : Not Ready among households that have pets.

This assumption allows us to find the percentage of households that have pets that aren't ready for a pet from data on pet abandonment. There are a total of 37 million households with cats as of 2020.^[21] In addition, about 1.8 million households in the U.S. surrender their pet to a pet shelter per year.^[22] Assuming this accounts for all not ready households in a year, we find a ratio of about 95% : 5% of households with pets that are ready and not ready. From this, we can find

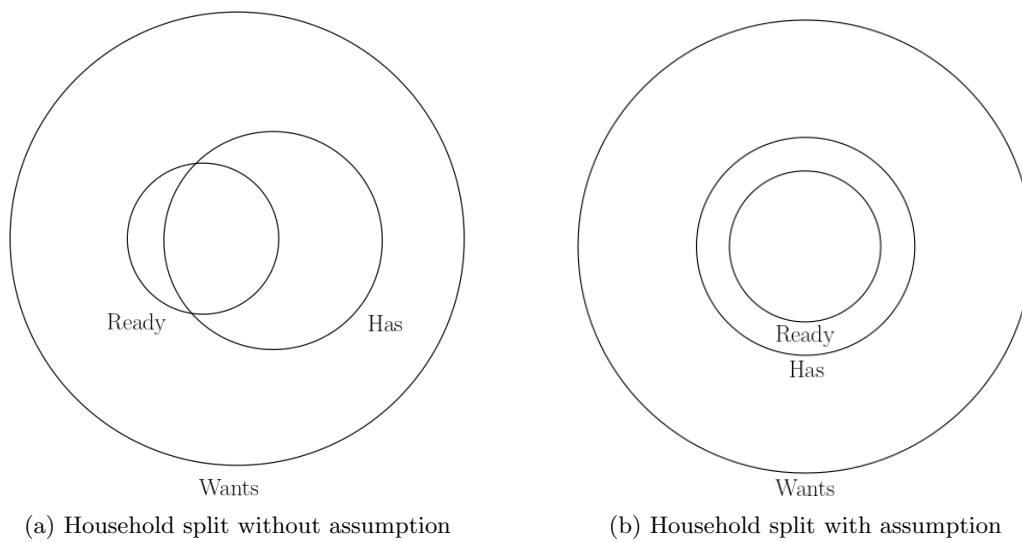


Figure 6: Usefulness of new assumption to finding pet retention

a conversion factor, k , between the readiness our model outputs to the readiness of households that have that pet.

$$\frac{29.2}{70.8}k = \frac{95}{5} = 19$$

$$k = 19 \cdot \frac{70.8}{29.2} \approx 46.1.$$

We take advantage of our previously developed model to reduce the amount of data we need to find on abandonment in all countries. More accurate approximations could be obtained with more data, but given a lack of data we assume our model is a good enough predictor of readiness.

5.1.2 Pet Ownership (Stochastic Differential Equations)

Table 13. Additional Definitions for Task 3

Variable/Function	Definition
$f_D(N_t, t)$	Deterministic change of N_t
$f_S(N_t, t)$	Stochastic component of change
$\mu(t)$	Hazard Function
$H(h) = \int_0^h \mu(t)dt$	Cumulative Hazard Function from year 0 to h
$LDF(t) = \mu(t)e^{-H(t)}$	Lifetime Density Function
$CLDF(h) = \int_0^h LDF(t)dt$	Cumulative Lifetime Density Function
N_t	Number of households with a pet at time t
a	Coefficient representing growth of N_t per year
b	Coefficient for death of pets in households per year
c	Coefficient for randomness / unpredictability of growth
M	Max pet age

Due to the lack of data regarding pet ownership over time for less common pets in each country, we decided to implement a probabilistic element to our projection model. We decided to use a **Stochastic Differential Equation** (SDE) to model pet ownership growth. The general form of a SDE implements both deterministic and probabilistic elements:

$$dN_t = f_D(N_t, t)dt + f_S(N_t, t)dW_t$$

In this equation, $f_D(N_t, t)$ is a deterministic function and $f_S(N_t, t)$ is probabilistic. The term dW_t represents the Wiener process or Brownian motion, essentially providing a randomness factor to the change of N_t . It follows a normal probability distribution (defined in Section 3.3.2) with mean 0 and variance t , allowing it to model unpredictability or chaos in dynamic systems. We choose relatively simple functions

$$\begin{aligned} f_D(N_t, t) &= a - bN_t \\ f_S(N_t, t) &= c \end{aligned}$$

Where a represents a standard growth rate for pet population, bN_t represents death rate of pets, and c represents a volatility term that is used for randomness. We will choose $c = .01$ so that probability has a reasonable impact but not enough to completely change the trend of growth.

To determine b , we have to find the rate of death for pets in a year. We will use a model that was created for human death, but adapt it for the case of pets. We use the lifetime density function (LDF) defined in Table 13:

$$LDF(t) = \mu(t)e^{-H(t)}.$$

In order to use this, we have to figure out a hazard function that fits the situation. The hazards that humans live through change drastically throughout their lives, but, generally, pets with homes live protected lives. For this reasoning, we choose a linear $\mu(t) = \frac{t}{k}$ that represents that the overall hazard of a pet's life increases only due to aging effects. We will choose a k that aligns with existing data on mean death age for pets. With $\mu(t) = \frac{t}{k}$, we get

$$H(h) = \int_0^h \frac{t}{k} dt = \frac{1}{k} \left(\frac{h^2}{2} - 0 \right) = \frac{h^2}{2k}.$$

For this problem, we consider $h = 5, 10$, and 15 . The integral of the LDF from x to $x + \Delta t$ with respect to time returns a probability of death given a current age x and a time length h , in years. So, the probability that a pet of age x will live to $x + \Delta t$ years is

$$\int_x^{x+\Delta t} LDF(t)dt = \int_x^{x+\Delta t} \frac{t}{k} e^{-\frac{t^2}{2k}} dt = -e^{-\frac{(x+\Delta t)^2}{2k}} - e^{-\frac{x^2}{2k}}.$$

This is also the cumulative lifetime density function (CLDF) from x to $x + \Delta t$. Finally, we have to account for the fact that there are many pets of different ages for each year to find b , representing the rate of death of the pets. For a max age, M , we integrate over all ages and take the average to find the average probability of death:

$$b = \frac{1}{M} \int_0^M \left(\int_x^{x+\Delta t} LDF(t)dt \right) dx = \frac{1}{M} \int_0^M \left(\int_x^{x+\Delta t} \frac{t}{k} e^{-\frac{t^2}{2k}} dt \right) dx$$

There is no closed form for this double integral, so we will be using computing technology to directly compute it. Solving for b here is dependent on how we obtain N_t : due to the nature of a stochastic differential equation, we use the Euler-Maruyama method of successive numerical approximation instead of finding a solution to the differential equation. As such, we set Δt to some small constant to approximate continuous growth. Then, given M , we find k analytically by analyzing when the cumulative death percentage matches a mean death age for a pet.

Finally, the coefficient a is found by data analysis on current demographic trends. We will show a worked example for cats in the U.S. We take the maximum age of a cat to be 38 years (the oldest recorded pet cat age), then solve for b . We take $k = 110$ since the $CLDF(30)$ returns close to 1 and $CLDF(15)$, the average pet cat age, is around 70%. This gets us $b = .0002628$, so about .026% of the cat population in the U.S. dies per year. The percentage of US households with cats increased by 4% (25% to 29%) over six years (2016 to 2022).^[22] This results in a yearly change of 0.8752 million households per year, so we take $a = 0.8752$. So, the SDE for pet cats in the U.S. is:

$$dN_t = (0.8752 - .0002628N_t)dt + .01dW_t.$$

In 2024, the number of households with pet cats is 46.5 million. We set $N_0 = 46.5$ and run the Euler-Maruyama approximation (code provided in Appendix A). Our model projects that there will be 50.7 million pet cats in 2029, 55.3 million pet cats in 2034, and 59.642 million pet cats in 2039. For these, we know that 5% of the total were abandoned that year. This means that 2.5 million pet cats will be abandoned in 2029, for a 95.3% retention rate, 2.77 pet cats will be abandoned in 2034 for a 95.2% retention rate, and 2.98 pet cats will be abandoned in 2039 with a retention rate of 95.3% that year.

5.2 Full Application of Model

We utilized Model 1 and the strategies implemented in 1b to determine readiness for each new species for each country. Using the conversion constant we calculated in 5.1.1 ($k = 46.1$), we are able to convert the ratio of Ready to Not Ready households in the whole country to the Ready to Not Ready ratio among households in each country with those pets. Given a percent readiness for country, p , we compute percent readiness of households with pets (q) with

$$46.1 \cdot \frac{p}{100 - p} = \frac{q}{100 - q}.$$

Table 14. Percent readiness of entire country and Percent readiness of households with pets per country

	U.S.		U.K.		Japan	
	Whole	Have pet	Whole	Have pet	Whole	Have pet
Cat	29.2%	95%	41.2%	97.0%	36.2%	96.3%
Dog	13.2%	87.5%	16.1%	89.8%	15.7%	89.6%
Parakeet	35.5%	96.2%	41.2%	97.0%	45.7%	97.5%
Goldfish	45.9%	97.5%	52.2%	98.1%	59.8%	98.6%
Boa	45.9%	97.5%	52.1%	98.0%	59.5%	98.5%

Then, we calculate the SDF for each species in each country. We keep the stochastic coefficient $c = .01$ the same for all SDF's, b changes per species (using M) and a changes per country and species. A table with every a and b value for the SDF's found can be found in Appendix E.

Table 15. Number of Households with Pets, in Millions, Years after 2024

	Initial			U.S.		
	US	UK	Japan	5 Yrs	10 Yrs	15 Yrs
Cat	46.5	7.332	5.36	50.7	55.3	59.64
Dog	65.5	8.742	6.481	72.54	80.01	87.62
Parakeet	6.1	.649	.838	6.26	6.42	6.59
Goldfish	11.1	2.8	2.682	11.50	11.91	12.25
Boa	3.55	.395	.2793	3.62	3.69	3.77
	U.K.			Japan		
	5 Yrs	10 Yrs	15 Yrs	5 Yrs	10 Yrs	15 Yrs
Cat	8.21	9.06	10.07	5.30	5.25	5.21
Dog	9.74	10.72	11.80	5.45	4.37	3.29
Parakeet	0.768	0.879	0.992	0.837	0.833	0.826
Goldfish	2.01	1.22	0.44	2.64	2.64	2.59
Boa	0.394	0.393	0.391	0.277	0.278	0.276

Retention rates were gathered as a percentage of pets not abandoned to pets relocated in total. That is,

$$\text{Retention} = 1 - \frac{\# \text{Households Abandoning Pets}}{\# \text{New Households with pets}}$$

Rates were around the same for each year, and here are the full results:

Table 16. Yearly Retention Rate of Households with Pets

	U.S.	UK	Japan
Cat	95.3%	97.1%	96.4%
Dog	88.8%	90.7%	90.6%
Parakeet	96.3%	97.1%	97.6%
Goldfish	97.6%	98.1%	98.6%
Boa	97.6%	98%	98.5%

6 Sensitivity Analysis

6.1 HR-SPLIT Model

6.1.1 Strengths

Our HR-SPLIT model utilizes a comprehensive approach to assessing household readiness for pet ownership, accounting for a variety of critical factors such as financial readiness, space availability, time commitment, proximity to veterinary services, and knowledge level. This holistic approach enables a nuanced evaluation beyond simple yes/no criteria, leading to better outcomes for pets and owners. In addition, users are able to enter information readily available to them to determine their readiness. For cats and dogs, our model also provides the valuable input of whether or not that family should buy pet insurance (if they don't have it already).

In terms of the method we used, creating HR-SPLIT allowed us to combine the factors into one readiness index, using a geometric mean, with a preserved mean of 0.5. This allowed us to easily split readiness into categories that tell the user elements to improve on before getting a pet even if they are ready, making sure pets live healthy and fun-filled lives. A specific strength of the model is that we interpreted each model individually, avoiding the need of weighing factors.

Though we were limited to 10 factors, more factors would increase the reliability of HR-SPLIT, ensuring that families are absolutely ready. Some of these are embedded in our assumptions, for example, that a household is aware of any pet policies that is imposed on them. HR-SPLIT is easily extendable to more factors, splitting each factor into more subregions, and more, making this model able to incorporate new factors and updates.

6.1.2 Weaknesses

The main issue with this model is that it is data-dependent. Though the user input is simple to get on an individual scale, when determining readiness for entire countries in Task 1b, we were faced with how much data our model actually required.

A major assumption we made was that all data provided by the user was factual. These factors can greatly vary between individuals' perceptions and reality. If we did not make this assumption, we would have to come up with a way to address this and calculate confidence intervals for when the data is true. For example, our knowledge level input was a piece of data that was very subjective. The user would not really know their exact knowledge level on a scale of 1 through 10. Techniques such as fuzzy logic, which involves dealing with reasoning that is approximate rather than fixed, or probabilistic models that incorporate user input variability, would have to be used to allow the model to account for the inherent uncertainty and subjectivity in certain inputs, providing a more nuanced assessment that reflects potential variations in input accuracy.

Further, an assumption our model made was that the household wants the pet. While this seemed reasonable when we first said it, it became an issue in determining which households in a country actually want a pet. This statistic is virtually impossible to find without a nationwide survey, so it required us to make workarounds in our interpretation of this assumption as we moved on to the next model.

6.2 SDE Model

6.2.1 Strengths

A major strength of using stochastic differential equations (SDEs) is their probabilistic aspect. Since data was limited, we have no way of completely determining growth rates as a function. We attempted to fit logistic curves to data until we realized that it did not work for some datasets and otherwise would require us to have specific data on each country for

each species. Implementing SDEs allows us to cover this ground by only requiring a growth rate (using data, but much less than otherwise needed) and death rate determined per species. Using a probabilistic approach reduced the need for simplifying assumptions in creating a demographic projection model.

Another strength of this model is how the death rate is calculated. The lifetime density function we used has been verified and tested with other methods by a research paper. Using a reasonable hazard function, we were able to determine a death rate to use in the SDE. This is much more reliable than coming up with arbitrary death factors based on data, especially due to the lack of data.

SDE models, similarly to HR-SPLIT, are highly receptive to changes in factors considered. We use variables as simple representations of growth, death, and stochastic factors, but each can be broken up and refined into clearer and more precise elements contributing to the model. Our model provides a glimpse into the power of SDEs and utilizes them effectively to predict pet demographics.

6.2.2 Weaknesses

We make a major assumption regarding pet retention: everyone who wants a pet and is ready for a pet gets a pet. Now, this obviously is not true, but it greatly simplified the process of making the model. A model not using this assumption would be more accurate to population data, but it would also require more data, which is especially problematic because that would also require knowing the number of households that want pets.

In all, this model is also only counting the number of households per country with or without pets. The total number of pets can be extrapolated by data, but it is not what our model is measuring. One reason for the abandonment of pets is when a pet gives birth and its owners cannot take care of the entire litter, so they bring them to a shelter. A situation like this isn't necessarily considered in our model, since that household will still have the status of having a pet even though an entire litter was abandoned in the process. To deal with this issue, a similar model can be created based on pet population data instead of households with pets. Death and stochastic elements would stay the same since it is also per pet.

Finally, though this model was built to compensate for a lack of data, it still suffers from that fact. It is hard to get abandonment rates per species per country, so we instead used a conversion factor between country readiness and house with pets readiness from our previous model's predictions. This means that the reliability of the data from this model also relies on the reliability of the first model, which is not ideal. In addition, finding data for growth rates itself was also an issue, but we were able to make do with the usefulness of the SDE.

7 Conclusion

Though humans and pets have coexisted for thousands of years, mistreatment and abandonment of pets persist. Households must be able to make educated decisions when buying pets, realizing that they are beings just as much as we are and have their own needs. Until the world realizes the true meaning of coexistence, we must keep pushing towards a society that is absent of intentional and unintentional maltreatment of pets. The models we have developed in this paper were made with the purpose of increasing awareness of the people, so that only those who are ready to care for pets will get them, and those who are not ready will have to wait.

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9 Appendix

Picking the Perfect Pet

Household Readiness: Are you ready to support a pet?

What pet are you looking for?

☐ Cat
 ☐ Parakeet
 ☐ Dog
 ☐ Goldfish
 ☐ Red-Tail Boa

How much floor space is available (m²)?

Enter Here:

How much vertical space is available (m)?

Enter Here:

How much money can you spend on your pet annually (USD)?

Enter Here:

Do you have a pet? If so, how well do they get along with other pets?

☐ N/A
 ☐ 0
 ☐ 1
 ☐ 2
 ☐ 3
 ☐ 4
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 8
 ☐ 9
 ☐ 10

Horribly Can co-exist Perfectly

How much free time do you have among all household members (hours)?

Enter Here:

What type of area is your household in?

☐ Urban
 ☐ Rural
 ☐ Suburban

How far is the closest veterinarian?

Enter Here:

How long are you willing to support your pet?

Enter Here:

How much do you know about this pet?

☐ 0
 ☐ 1
 ☐ 2
 ☐ 3
 ☐ 4
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 8
 ☐ 9
 ☐ 10

Absolutely Nothing Enough Everything

Figure 7: Interface of our Complete Household Readiness Model

9.1 A. Code

9.1.1 Model 1 Code

```

1 public class Main {
2     public static double floorSpace(double x, double FS) {
3         if (x >= 3 * FS) {
4             System.out.println("FS: " + 1);
5             return 1;
6         } else if (3 * FS > x && x >= 2 * FS) {
7             System.out.println("FS: " + 2 / 3.0);
8             return 2 / 3.0;
9         } else if (2 * FS > x && x >= FS) {
10            System.out.println("FS: " + 1 / 3.0);
11            return 1 / 3.0;
12        } else {
13            System.out.println("FS: " + 0);
14            return 0;
15        }
16    }
17
18    public static double verticalSpace(double x, double VS) {
19        if (x >= 1.5 * VS) {
20            System.out.println("VS: " + 1);
21            return 1;
22        } else if (1.5 * VS > x && x >= VS) {
23            System.out.println("VS: " + 1 / 2.0);
24            return 1 / 2.0;
25        } else {
26            System.out.println("VS: " + 0);
27            return 0;
28        }
29    }
30
31    public static double cost(double x, double C) {
32        if (x >= 2 * C) {
33            System.out.println("Cost: " + 1);
34            return 1;
35        } else if (x < C) {

```

```
36         System.out.println("Cost: " + 0);
37         return 0;
38     } else {
39         for (int i = 0; i < 10; i++) {
40             if ((1 + i / 9.0) * C > x && x >= (1 + (i - 1) / 9.0) * C) {
41                 System.out.println("Cost: " + i / 10.0);
42                 return i / 10.0;
43             }
44         }
45     }
46     System.out.println("Nothing");
47     return 0;
48 }
49
50 public static double freeTime(double x, double FT) {
51     if (x >= 5 * FT) {
52         System.out.println("FT: " + 1);
53         return 1;
54     } else if (x < FT) {
55         System.out.println("FT: " + 0);
56         return 0;
57     } else {
58         for (int i = 0; i < 5; i++) {
59             if ((i + 1) * FT > x && x >= i * FT) {
60                 System.out.println("FT: " + i / 5.0);
61                 return i / 5.0;
62             }
63         }
64     }
65     System.out.println("Nothing");
66     return 0;
67 }
68
69 public static double vet(String area, double x) {
70     if (area.equals("rural")) {
71         if (x <= 9) {
72             System.out.println("Vet: " + 1);
73             return 1;
74         } else if (9 < x && x <= 12) {
75             System.out.println("Vet: " + 2 / 3.0);
76             return 2 / 3.0;
77         } else if (12 < x && x <= 15) {
78             System.out.println("Vet: " + 1 / 3.0);
79             return 1 / 3.0;
80         } else {
81             System.out.println("Vet: " + 0);
82             return 0;
83         }
84     } else if (area.equals("suburban")) {
85         if (x <= 2) {
86             System.out.println("Vet: " + 1);
87             return 1;
88         } else if (2 < x && x <= 6) {
89             System.out.println("Vet: " + 2 / 3.0);
90             return 2 / 3.0;
91         } else if (6 < x && x <= 10) {
92             System.out.println("Vet: " + 1 / 3.0);
93             return 1 / 3.0;
94         } else {
95             System.out.println("Vet: " + 0);
96             return 0;
97         }
98     } else if (area.equals("urban")) {
99         if (x <= 1) {
100             System.out.println("Vet: " + 1);
101             return 1;
102         } else if (1 < x && x <= 3) {
103             System.out.println("Vet: " + 2 / 3.0);
104             return 2 / 3.0;
105         } else if (3 < x && x <= 5) {
106             System.out.println("Vet: " + 1 / 3.0);
107             return 1 / 3.0;
108         } else {
```

```
109         System.out.println("Vet: " + 0);
110         return 0;
111     }
112 }
113 return 0;
114 }
115
116 public static double knowledge(double x) {
117     if (x >= 8) {
118         System.out.println("Know: " + 1);
119         return 1;
120     } else if (x < 4) {
121         System.out.println("Know: " + 0);
122         return 0;
123     } else {
124         for (int i = 1; i <= 4; i++) {
125             if (i + 4 > x && x >= i + 3) {
126                 System.out.println("Know: " + i / 5.0);
127                 return i / 5.0;
128             }
129         }
130     }
131     return 0;
132 }
133
134 public static double priorpet(double x){
135     //if N/A (no other pets), default input is 10
136     if(x >= 8){
137         System.out.println("Prior: " + 1);
138         return 1;
139     }
140     else if(x < 4){
141         System.out.println("Prior: " + "0");
142     }
143     else{
144         for(int i = 1; i <= 4; i++){
145             if(x >= i+3 && x < i+4){
146                 System.out.println("Prior: " + i / 5.0);
147                 return i / 5.0;
148             }
149         }
150     }
151     return 0;
152 }
153
154 public static double commit(double x, double TE){
155     if(x >= TE){
156         System.out.println("TE: " + 1);
157         return 1;
158     }
159     else if(x < 0.25 * TE){
160         System.out.println("TE: " + 0);
161         return 0;
162     }
163     else{
164         for(int i = 1; i <= 3; i++){
165             if(x >= (i / 4.0) * TE && x < ((i+1) / 4.0) * TE){
166                 System.out.println("TE: " + i / 4.0);
167                 return i / 4.0;
168             }
169         }
170     }
171     System.out.println("Nothing");
172     return 0;
173 }
174
175 public static String Final(double FS, double VS, double VD, String Loc, double IK, double C, double
176 FT, double TE, double PC, String species) {
177     //Determine readiness for a given species
178     String[] specieslist = {"cat", "dog", "goldfish", "parakeet", "boa"};
179     int j = 0;
180     for(int i = 0; i < specieslist.length; i++){
181         if (specieslist[i].equals(species)){
182             j = i;
183         }
184     }
185 }
```

```

181     }
182 }
183 Animal a;
184 if(j==0){
185     a = new Cat();
186 }
187 else if(j == 1){
188     a = new Dog();
189 }
190 else if(j==2){
191     a = new Goldfish();
192 }
193 else if(j==3){
194     a = new Parakeet();
195 }
196 else if(j==4){
197     a = new Boa();
198 }
199 else{
200     a = new Cat();
201 }
202 double val = Math.pow(floorSpace(FS, a.getFloor()) * verticalSpace(VS, a.getVert()) * vet(Loc,
203 VD) * cost(C, a.getCost()) * knowledge(IK) * freeTime(FT, a.getFree()) * commit(TE, a.getExp()) *
204 priorpet(PC), 1 / 8.0);
205 System.out.println("Value: " + val);
206 if(val == 0){
207     return "Not Prepared";
208 }
209 else{
210     if(j==0 || j==1){
211         if (C >= 1.5 * a.getIcost()) {
212             System.out.println("We recommend you get insurance based on the amount you are
213 willing to spend on your pet.");
214         }
215         else {
216             System.out.println("Based on the amount you are willing to spend on your pet,
217 we do not recommend you get pet insurance.");
218         }
219     }
220     if (0 < val && val <= 0.5) {
221         return "Somewhat Prepared";
222     }
223     if (0.5 < val && val < 1) {
224         return "Very Prepared";
225     }
226     if (val == 1) {
227         return "Fully Prepared";
228     }
229 }
230 return "";
231 }
232 public static String Catver1(double FS, double VS, double VD, String Loc, double IK, double C,
233 double FT) {
234     Animal a = new Cat();
235     double val = Math.pow(floorSpace(FS, a.getFloor()) * verticalSpace(VS, a.getVert()) * vet(Loc,
236 VD) * cost(C, a.getCost()) * knowledge(IK) * freeTime(FT, a.getFree()), 1 / 6.0);
237 System.out.println("Value: " + val);
238 if(val == 0){
239     return "Not Prepared";
240 }
241 else{
242     if (C >= 1.5 * a.getIcost()) {
243         System.out.println("We recommend you get insurance based on the amount you are
244 willing to spend on your pet.");
245     }
246     else {
247         System.out.println("Based on the amount you are willing to spend on your pet,
248 we do not recommend you get pet insurance.");
249     }
250     if (0 < val && val <= 0.5) {
251         return "Somewhat Prepared";
252     }
253     if (0.5 < val && val < 1) {

```

```

246         return "Very Prepared";
247     }
248     if (val == 1) {
249         return "Fully Prepared";
250     }
251 }
252 return "";
253 }
254
255 public static void main(String[] args) {
256     // double FS, double VS, double VD, String Loc, double IK, double C, double FT, double TE,
257     double PC, String species
258     //1a, very qualified
259
260     System.out.println(Catver1(20, 2.4, 3, "urban", 8, 2500, 3));
261     System.out.println(Catver1(75, 2.5, 7.8, "suburban", 7, 1900, 3));
262     System.out.println(Catver1(140, 2.7, 11, "rural", 6, 4000, 8));
263
264     //1a, somewhat qualified
265
266     System.out.println(Catver1(18, 2, 4, "urban", 6, 2300, 2));
267     System.out.println(Catver1(70, 2.6, 8, "suburban", 4, 1500, 5));
268     System.out.println(Catver1(110, 3, 15, "rural", 5, 1600, 3));
269
270     //1a, not qualified
271
272     System.out.println(Catver1(15, 2.3, 2, "urban", 7, 2000, 1));
273     System.out.println(Catver1(60, 2.4, 8, "suburban", 6, 1200, 4));
274     System.out.println(Catver1(150, 2.8, 23, "rural", 8, 2000, 3));
275
276     //2a, qualified
277
278     System.out.println(Final(10, 2.5, 1.5, "urban", 5, 3925, 3, 15, 10, "cat"));
279     System.out.println(Final(15, 2.8, 9, "suburban", 9, 2100, 3.5, 20, 7, "dog"));
280     System.out.println(Final(0.95, 1, 4, "urban", 4, 800, 1, 12, 10, "parakeet"));
281     System.out.println(Final(1, 0.75, 5, "suburban", 10, 1600, 1.5, 15, 9, "goldfish"));
282     System.out.println(Final(5, 1, 14, "rural", 6, 1900, 6, 20, 6, "boa"));
283
284     //2a, unqualified
285
286     System.out.println(Final(7, 2.2, 10, "urban", 6, 1600, 1.5, 12, 10, "cat"));
287     System.out.println(Final(10, 2, 7, "suburban", 5, 700, 5, 10, 10, "dog"));
288     System.out.println(Final(0.5, 1.2, 12, "rural", 10, 800, 3, 20, 9, "parakeet"));
289     System.out.println(Final(0.9, 0.6, 4, "suburban", 2, 700, 1, 4, 10, "goldfish"));
290     System.out.println(Final(3, 0.8, 5, "rural", 7, 1320, 2, 20, 1, "boa"));
291
292 }
293 }

```

```

1 public class Animal{
2     public double C;
3     public double FS;
4     public double VS;
5     public double FT;
6     public double TE;
7     public double CI;
8     public Animal(){
9         super();
10    }
11    public double getCost(){
12        return C;
13    }
14    public double getFloor(){
15        return FS;
16    }
17    public double getVert(){
18        return VS;
19    }
20    public double getFree(){
21        return FT;
22    }
23    public double getExp(){

```

```
24         return TE;
25     }
26     public double getIcost(){
27         return CI;
28     }
29 }
```

```
1 public class Cat extends Animal{
2     public Cat(){
3         super();
4         C = 1399.0;
5         FS = 2.0;
6         VS = 2.0;
7         FT = 1.25;
8         TE = 15.0;
9         CI = 1586.0;
10    }
11 }
```

```
1 public class Dog extends Animal{
2     public Dog(){
3         super();
4         C = 1762.0;
5         FS = 6.0;
6         VS = 2.0;
7         FT = 4.0;
8         TE = 13.0;
9         CI = 2171;
10    }
11 }
```

```
1 public class Parakeet extends Animal{
2     public Parakeet(){
3         super();
4         C = 264.0;
5         FS = 0.7;
6         VS = 0.5;
7         FT = 1.0;
8         TE = 15.0;
9     }
10 }
```

```
1 public class Goldfish extends Animal{
2     public Goldfish(){
3         super();
4         C = 374.0;
5         FS = 0.25;
6         VS = 0.5;
7         FT = 0.5;
8         TE = 20.0;
9     }
10 }
11 }
```

```
1 public class Boa extends Animal{
2     public Boa(){
3         super();
4         C = 520.0;
5         FS = 1.3;
6         VS = 0.7;
7         FT = 0.5;
8         TE = 25.0;
9     }
10 }
```

9.1.2 Euler-Maruyama Approximation Code (Task 3)

```
1 import java.util.Random;
2
3 public class EulerMaruyama {
4
5     public static void main(String[] args) {
```

```

6      //dog, cat, fish, bird snake
7      double[] usarates = {1.48784, 0.8752, 0.08194, 0.03826, 0.01533};
8      double[] ukrates = {0.2115, .188, -.1566, .0231, -0.0002558};
9      double[] jprates = {-.2133, -.003465, -.00174, -.000543, -.000181};
10
11     double[] usainit = {65.5, 46.5, 11.1, 6.1, 3.55};
12     double[] ukinit = {8.742, 7.332, 2.8, .649, .395};
13     double[] jpinit = {6.481, 5.36, 2.682, .838, .2793};
14
15     double[] drates = {.0002628, .0003311, .0003440, .0002269, .0002477};
16     //USA
17     for(int i = 0; i < usarates.length; i++){
18         taka(usainit[i], usarates[i], drates[i]);
19     }
20     //UK
21     for(int i = 0; i < usarates.length; i++){
22         taka(ukinit[i], ukrates[i], drates[i]);
23     }
24     //JP
25     for(int i = 0; i < usarates.length; i++){
26         taka(jpinit[i], jprates[i], drates[i]);
27     }
28 }
29 public static void taka(double init, double grate, double drate){
30     double dt = 0.01; // Time step size
31     double T = 15; // Final time (goes up to 15, can measure 5 and 10)
32     int N = (int) (T / dt); // Number of steps
33     Random rand = new Random();
34     double t5 = 0;
35     double t10 = 0;
36     double t15 = 0;
37     double[] X = new double[N + 1];
38     int n = 20;
39     X[0] = init;
40     for(int j = 0; j < n; j++){
41         for (int i = 0; i < N; i++) {
42             double dW = Math.sqrt(dt) * rand.nextGaussian(); // Sample from normal distribution
43             X[i + 1] = X[i] + (grate - drate * X[i]) * dt + .01 * X[i] * dW; // Euler-Maruyama
44         }
45         // for(int i = 1; i <= T; i++){
46             //System.out.println("X_" + i + " = " + X[100 * i]);
47         //}
48         t5 += X[500];
49         t10 += X[1000];
50         t15 += X[1500];
51     }
52     System.out.println("Averages: " + "\n5 years: " + t5 / n + " \n10 years: " + t10 / n + " \n15
years: " + t15 / n);
53 }
54 }

```

9.2 B. Specifics of Model 1 Functions

$MVD_T = 15, 10$, or 5 . If $MVD_T = 15$ (rural), we have

$$f_{Dist}(VD, 15) = \left\{ \begin{array}{ll} 1 & VD \leq 9 \\ \frac{2}{3} & 9 < VD \leq 12 \\ \frac{1}{3} & 12 < VD \leq 15 \\ 0 & 15 < VD \end{array} \right\}.$$

If $MVD_T = 10$ (suburban),

$$f_{Dist}(VD, 10) = \left\{ \begin{array}{ll} 1 & VD \leq 2 \\ \frac{2}{3} & 2 < VD \leq 6 \\ \frac{1}{3} & 6 < VD \leq 10 \\ 0 & 10 < VD \end{array} \right\}.$$

If $MVD_T = 5$ (urban),

$$f_{Dist}(VD, 5) = \begin{cases} 1 & VD \leq 1 \\ \frac{2}{3} & 1 < VD \leq 3 \\ \frac{1}{3} & 3 < VD \leq 5 \\ 0 & 5 < VD \end{cases}.$$

For floor space, we have:

$$f_{FS}(FS_H, FS_{pet}) = \begin{cases} 1 & 3FS_{pet} \leq FS_H \\ \frac{2}{3} & 2FS_{pet} \leq FS_H < 3FS_{pet} \\ \frac{1}{3} & FS_{pet} \leq FS_H < 2FS_{pet} \\ 0 & FS_H < FS_{pet} \end{cases}$$

For vertical height, we have:

$$f_{VS}(VS_H, VS_{pet}) = \begin{cases} 1 & 1.5VS_{pet} \leq VS_H \\ \frac{1}{2} & VS_{pet} \leq VS_H < 1.5VS_{pet} \\ 0 & VS_H < VS_{pet} \end{cases}$$

9.3 C. Complete Data and Graphs for 1b

Table 17. Salary & Percent Qualified in the U.S.

Salary Range	Number of Households	Percent Qualified Based on Salary
Less than \$5000	6,917,000	0%
\$5000 to \$9999	3,459,000	0%
\$10,000 to \$14,999	5,231,000	0%
\$15,000 to \$19,999	5,087,000	0%
\$20,000 to \$24,999	5,707,000	0%
\$25,000 to \$29,999	5,242,000	0%
\$30,000 to \$34,999	5,613,000	10.28%
\$35,000 to \$39,999	4,902,000	47.11%
\$40,000 to \$49,999	10,249,000	90.26%
\$50,000 to \$59,999	8,876,000	99.28%
\$60,000 to \$79,999	16,146,000	100%
\$80,000 to \$99,999	12,003,000	100%
Over \$100,000	39,071,000	100%

Table 18. U.S. Leisure time per income range

Weekly Salary	L_{total}	L_{other}	Mean	$1 - \text{cdf}(0, 1.25)$
\$0 to \$760	4.0857	2.94	1.1457	0.4585
\$761 to \$1,160	4.19	2.92	1.27	0.508
\$1,161 to \$1,810	4.30	3.117	1.183	0.4733
\$1,811 and higher	4.014	2.74	1.274	0.5096

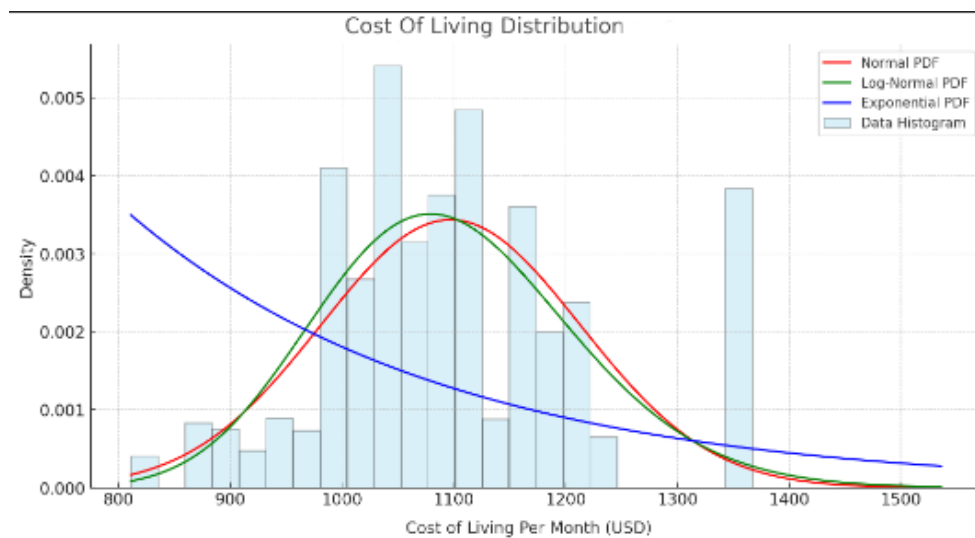


Figure 8: Distribution Data of Japan's Cost of Living

Table 19. Japan Salary And Percent Qualified Based on Salary

Salary Range	Number of Households	Percent Qualified Based on Salary
Less than \$6,645	3,832,400	0%
\$6,646 to \$13,290	7,436,000	26.7%
\$13,291 to \$19,935	8,351,200	99.8%
\$19,936 to \$26,580	7,264,400	100%
\$26,581 to \$33,226	5,891,600	100%
\$33,227 to \$39,871	4,804,800	100%
\$39,872 to \$46,516	4,175,600	100%
\$46,517 to \$53,161	3,546,400	100%
\$53,162 to \$59,806	2,802,800	100%
\$59,807 to \$66,452	2,059,200	100%
\$66,453 to \$73,097	1,773,200	100%
\$73,098 to \$79,742	1,201,200	100%
\$79,743 to \$86,387	972,400	100%
\$86,388 to \$93,032	629,200	100%
\$93,033 to \$99,678	572,000	100%
Over \$99,679	2,059,200	100%

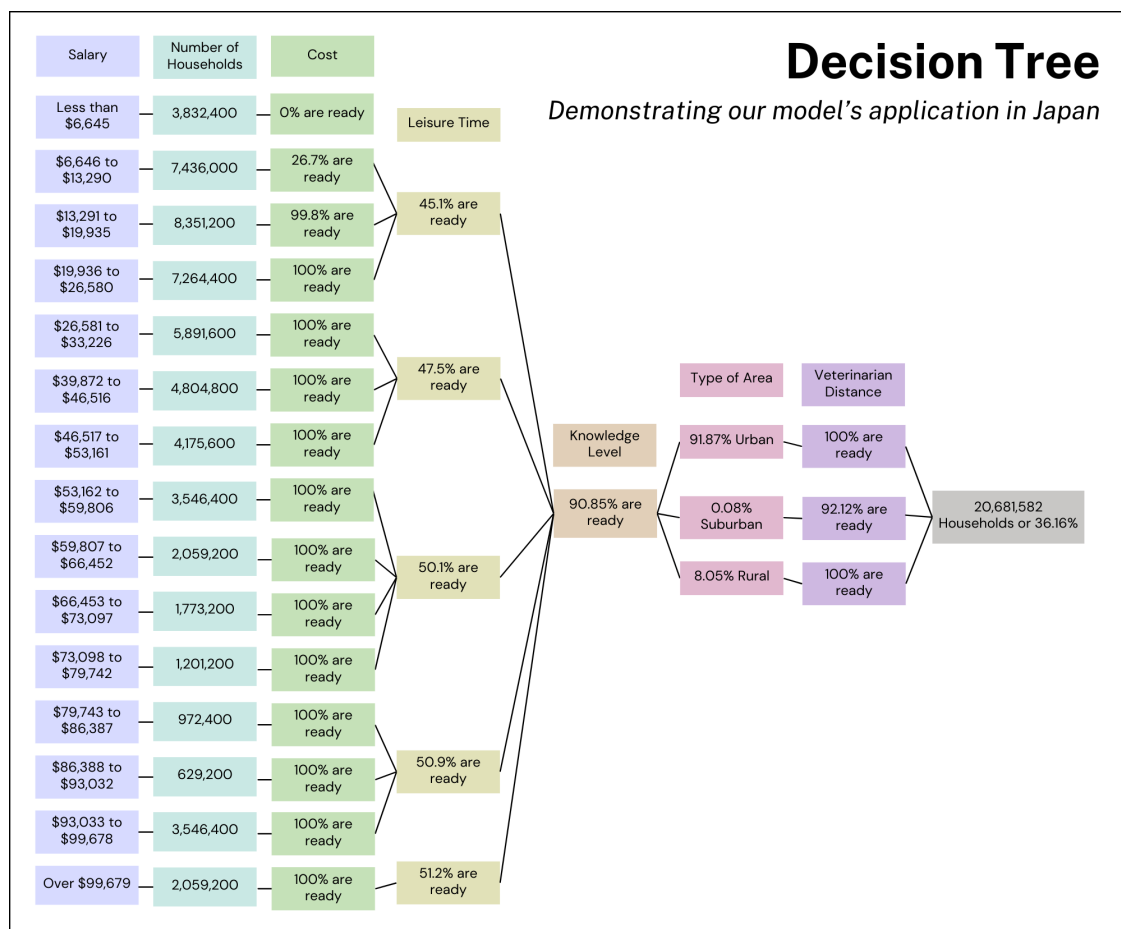


Figure 9: Decision Tree for Japan

Table 20. U.K. Salary And Percent Qualified Based on Salary

Salary Range	Number of Households	Percent Qualified Based on Salary
Less than \$13,185	1,692,000	0%
\$13,186 to \$26,305	5,640,000	0.953%
\$26,306 to \$39,490	5,358,000	99.9%
\$39,491 to \$52,676	3,948,000	100%
\$52,677 to \$65,861	3,102,000	100%
\$65,862 to \$79,047	2,256,000	100%
\$79,048 to \$92,232	1,692,000	100%
\$92,233 to \$105,418	1,128,000	100%
\$105,419 to \$118,603	846,000	100%
\$118,604 to \$131,789	564,000	100%
Over \$131,790	1,974,000	100%

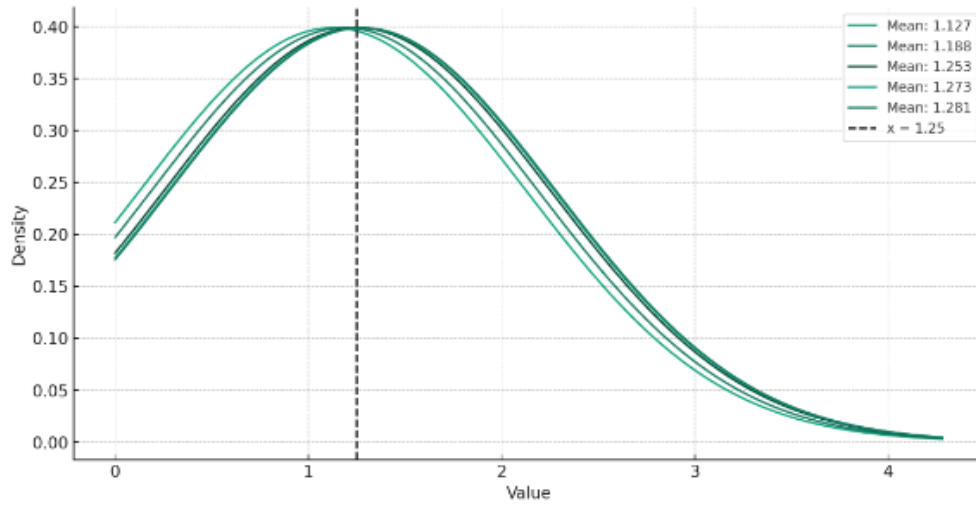


Figure 10: Normal Curves for Leisure Data of Japan at Different Income Ranges

9.4 D. Species Cost Calculations

By [10], the average annual cost for dogs is \$1391 with first-year additional cost of \$1830. Assuming a once in a lifetime accident costing \$3000 and an average lifespan of 13 years across dogs of all sizes, we have

$$C_{dog,N} = 1391 + 1830 \left(\frac{1}{13} \right) + 3000 \left(\frac{1}{13} \right) = \$1762.$$

Or, with insurance, from [13], complete dog insurance is \$640 per year. This gets

$$C_{dog,Y} = 1391 + 1830 \left(\frac{1}{13} \right) + 640 = \$2171.$$

By [25], Parakeets cost \$250 annually with an additional \$215 in the first year, not accounting for emergency costs. Since pet parakeets stay in a cage for most of their lives, we will assume no emergency bills incur for the minimum possible annual cost. For a lifespan of 15 years, this comes out to

$$C_{parakeet} = 250 + 215 \left(\frac{1}{15} \right) = \$264.$$

Goldfish cost \$360 annually with an additional \$275 in the first year. Again, assuming no accidents and a 20 year life span, we have

$$C_{goldfish} = 360 + 275 \left(\frac{1}{20} \right) = \$374.$$

A Red-Tail Boa costs \$500 annually with an additional \$500 in the first year. With a 25 year lifespan, assuming no emergency costs, we get

$$C_{Boa} = 500 + 500 \left(\frac{1}{25} \right) = \$520.$$

9.5 E: SDF Coefficient Tables

Coefficients for a were found using existing growth trend data on households with pets.

Table 21. SDF Coefficient a

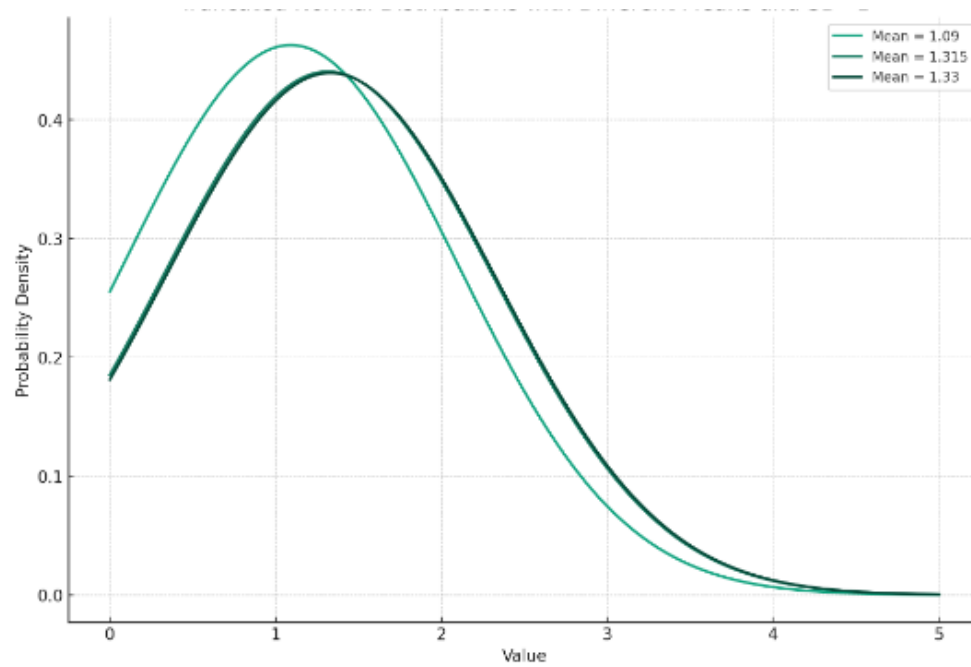


Figure 11: Normal Curves for Leisure Data of U.K. at Different Income Ranges

	U.S.	UK	Japan
Cat	0.8752	.188	-.003465
Dog	1.48784	0.2115	-.2133
Parakeet	0.03826	.0231	-.000543
Goldfish	0.08194	-.1566	-.00174
Boa	0.01533	-0.0002558	-.000181

Values for M taken from sources on oldest recorded age of each species as pets/in captivity.

Table 22. SDF Coefficient b (and Additional Info)

	Cat	Dog	Parakeet	Goldfish	Boa
b	.0002628	.0003311	.0003440	.0002269	.0002477
k	110	90	70	150	170
M	38	30	29	44	40
Avg Lifespan	15	13	15	20	25

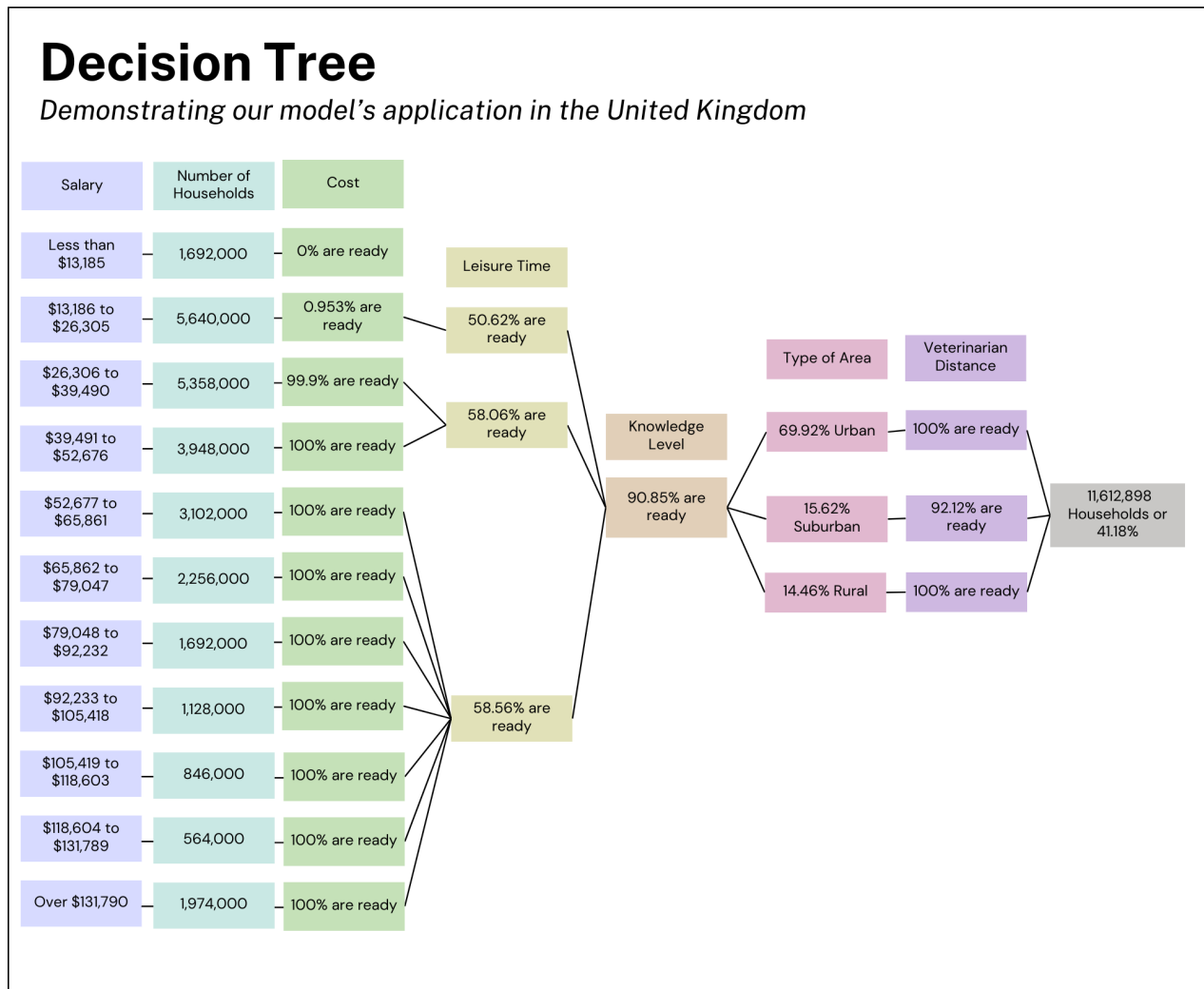


Figure 12: Decision Tree for the United Kingdom